Manipulation Planning with Convex Decompositions of C-free and NLP-Sampling

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THE UNIVERSITY OF EDINBURGH

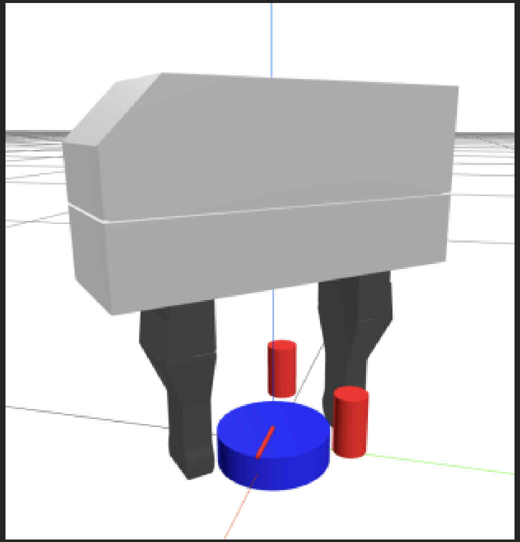
Edinburgh Centre for Robotics

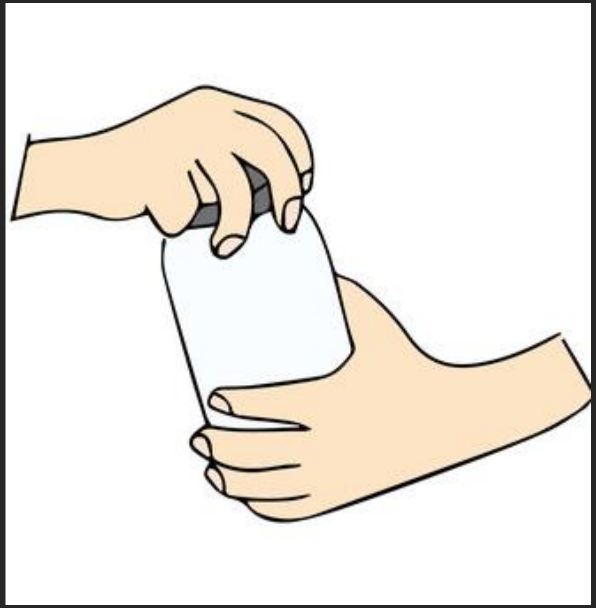
What is Planning?

What is Planning?

Finding a sequence of *actions* that solve a task.

Our Task: Opening a Bottle Cap





Where do I grasp and release the cap?

How much can I turn my wrist?

How many grasping actions do I need?

What if I put some obstacles?

And the minimum grasping actions?

How much can we turn (where can we reach) with each grasping action?

Can we concatenate grasping actions optimally?

Reachability in Geometric Motion and Manipulation Planning

Approaches to Motion Planning

Sampling-based methods

+ Simple. Global planning
+ Work well in lower dimensions
+ Can be massively accelerated through large-scale parallelization

- Scale poorly to high dimensions
- Memory inefficient
- Suboptimal plans and long planning times

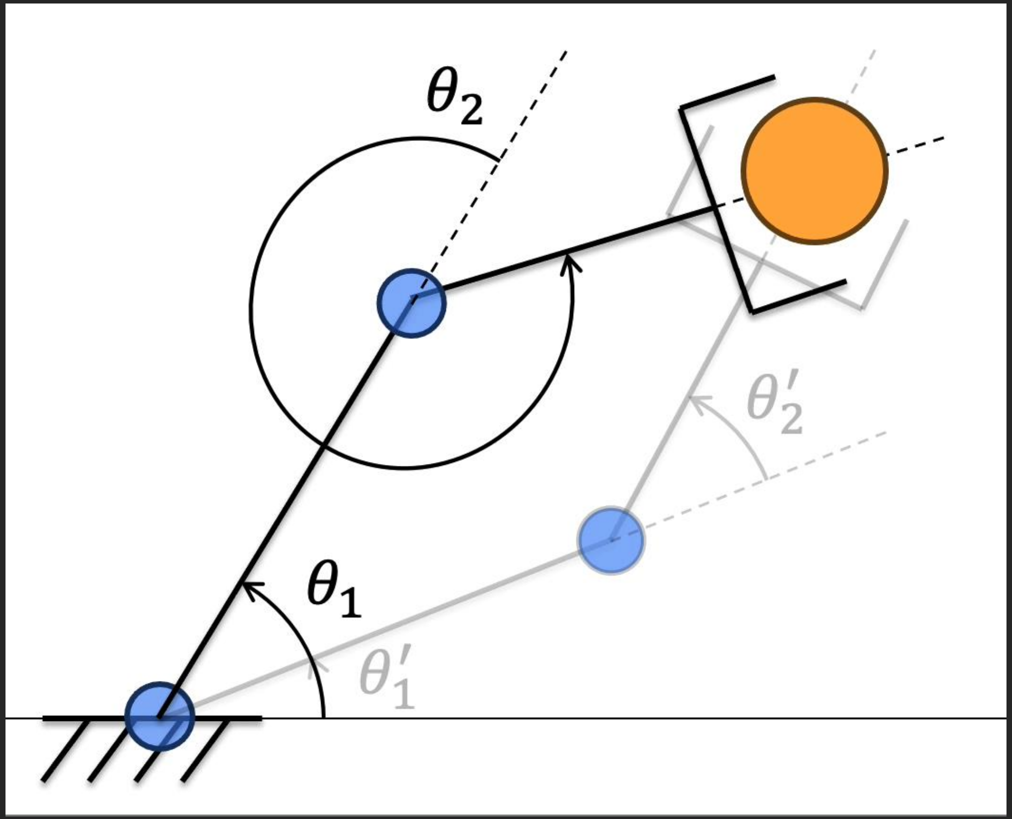
Nonlinear Trajectory Optimisation

+ Scales to higher dimensions
+ Allows for wider variety of costs and constraints

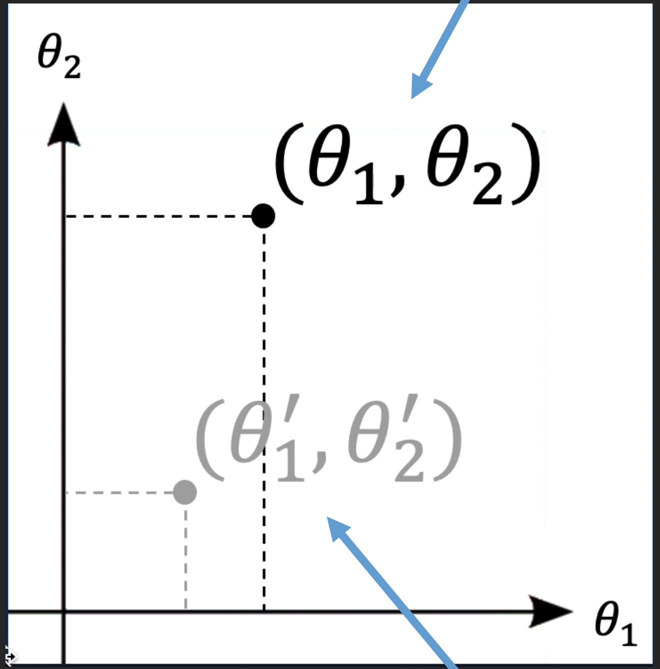
- Local optimization approaches needed
- Sensitive to initialization
- May struggle to find a feasible solution when one exists

Novel algorithms try to bridge the benefits of both.

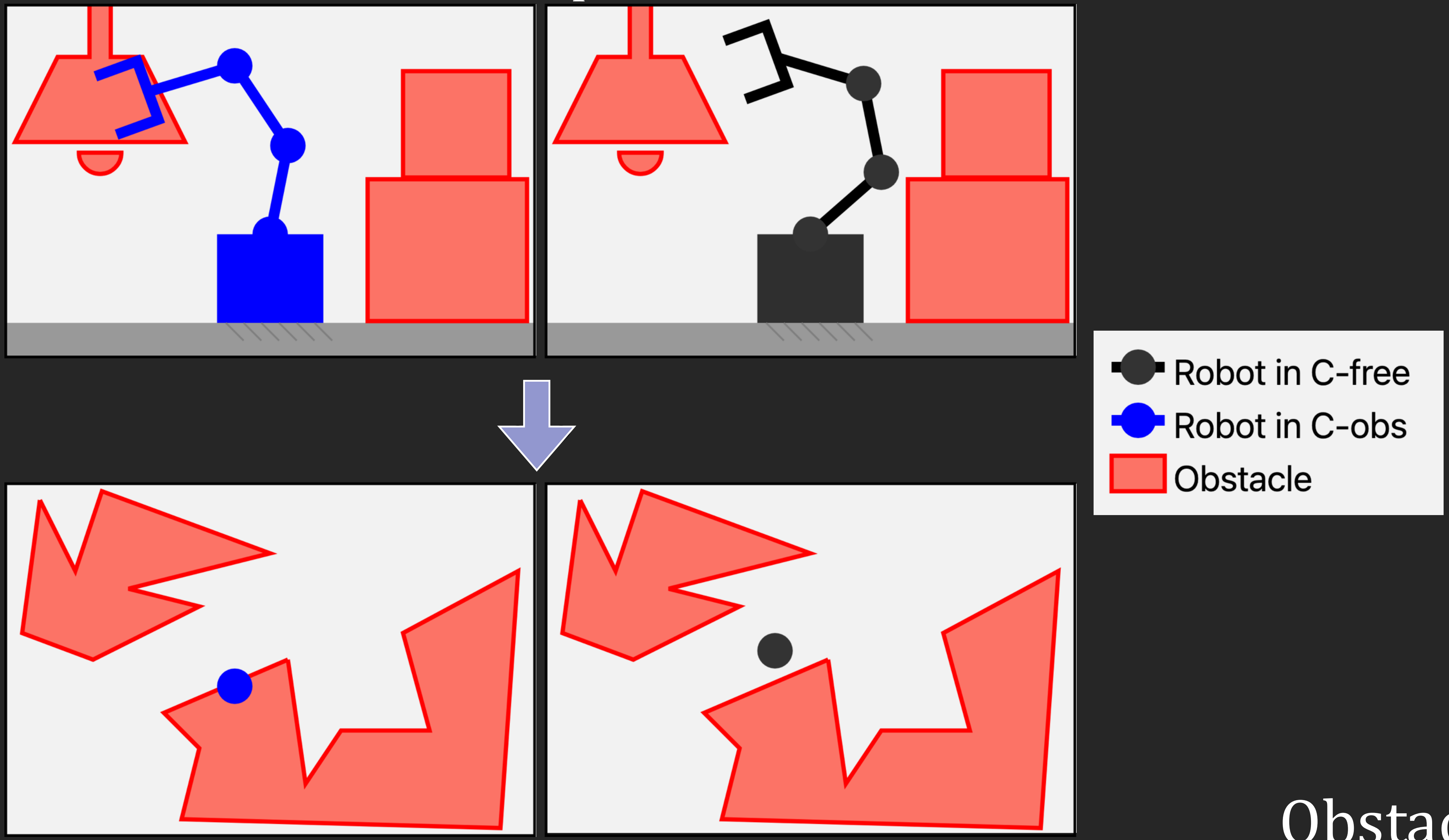
Task Space



Configuration Space

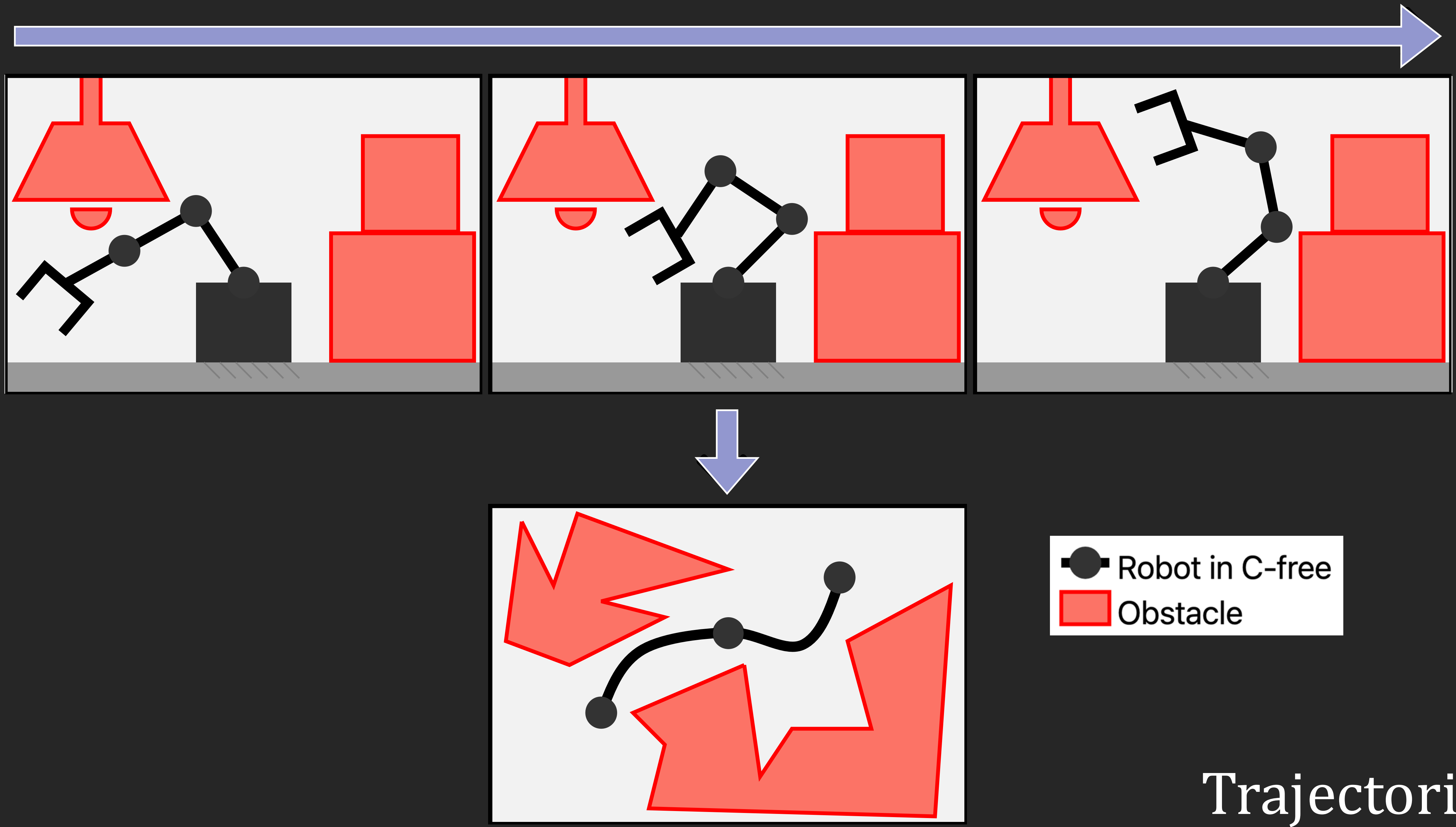


Task Space



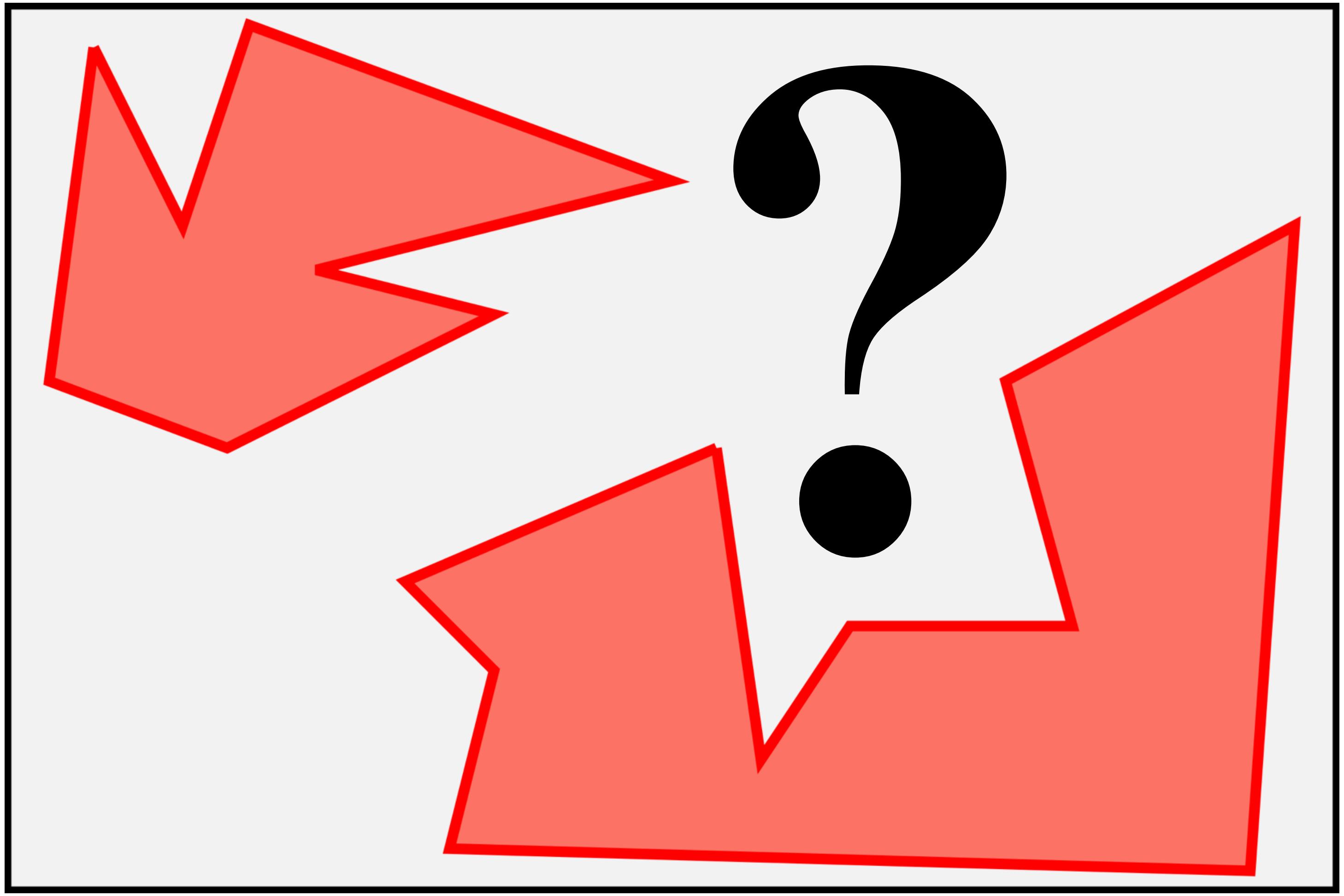
(Abstract) Configuration Space

Obstacles



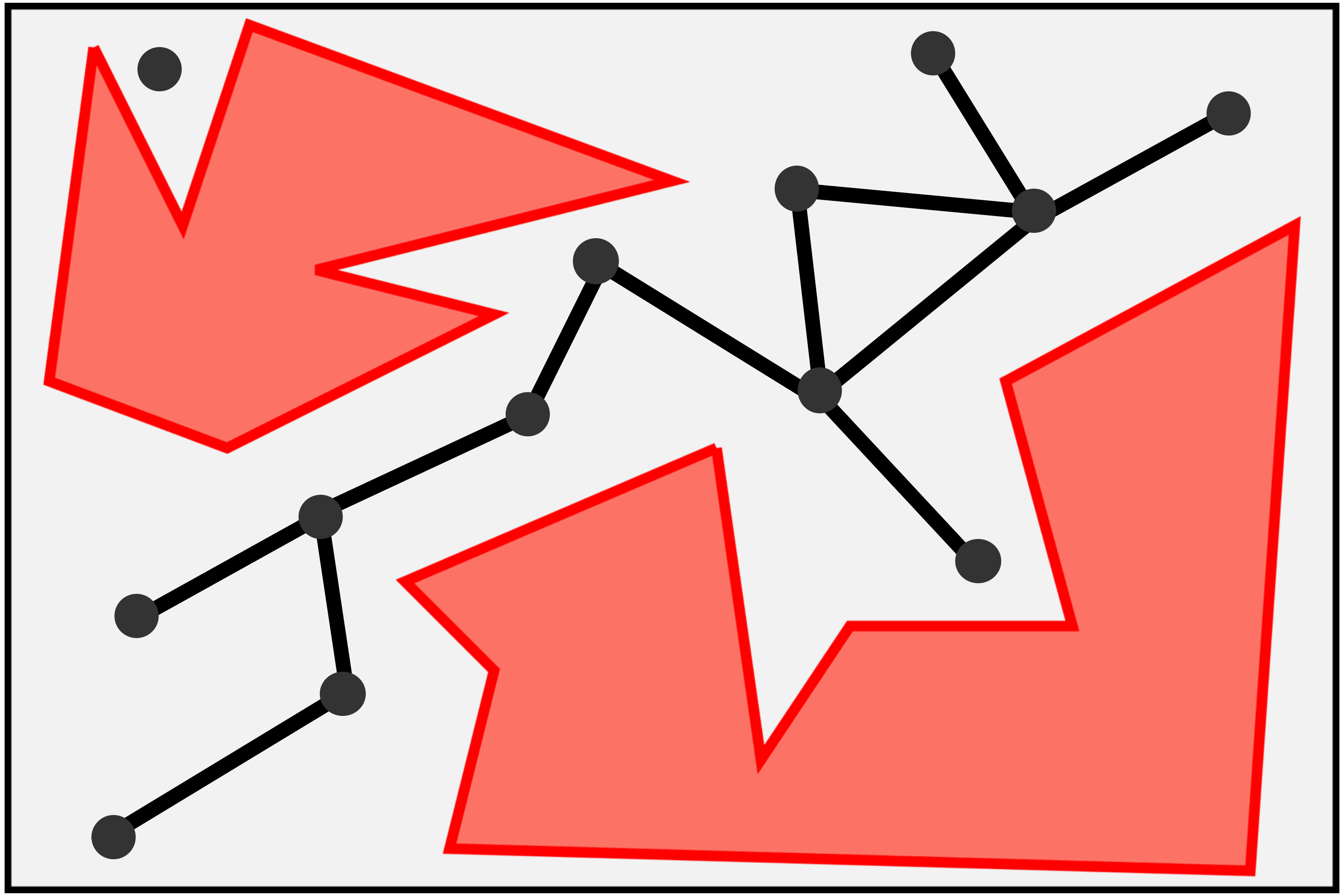
Trajectories

Collision-Free Configuration Space (C-free)



- How can we describe C-free?

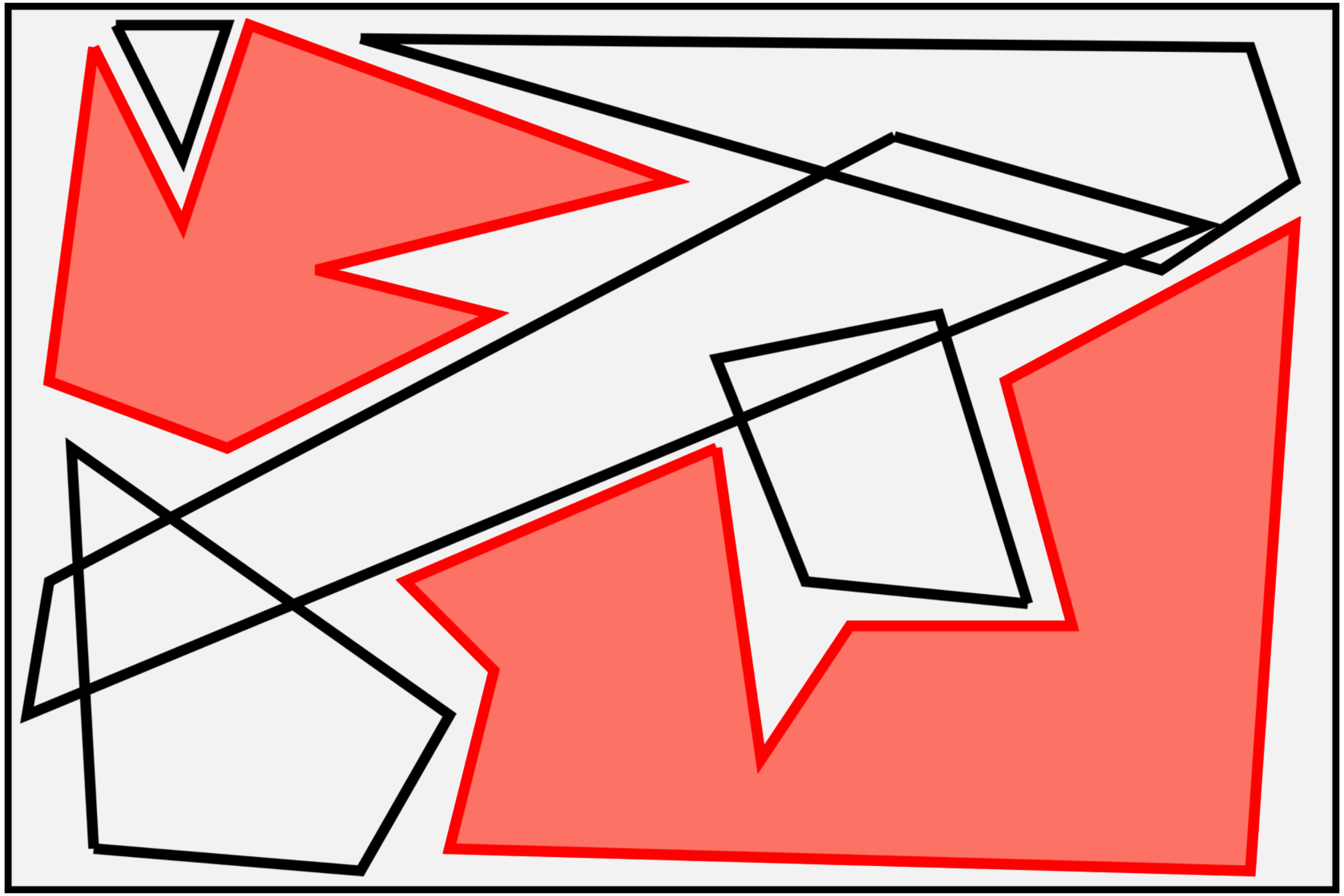
Collision-Free Configuration Space (C-free)



- How can we describe C-free?

1. **Sampling-Based Methods**
    - Probabilistic guarantees of non-collision
    - Probabilistic guarantees of completeness
    - Not inherently optimal

Collision-Free Configuration Space (C-free)

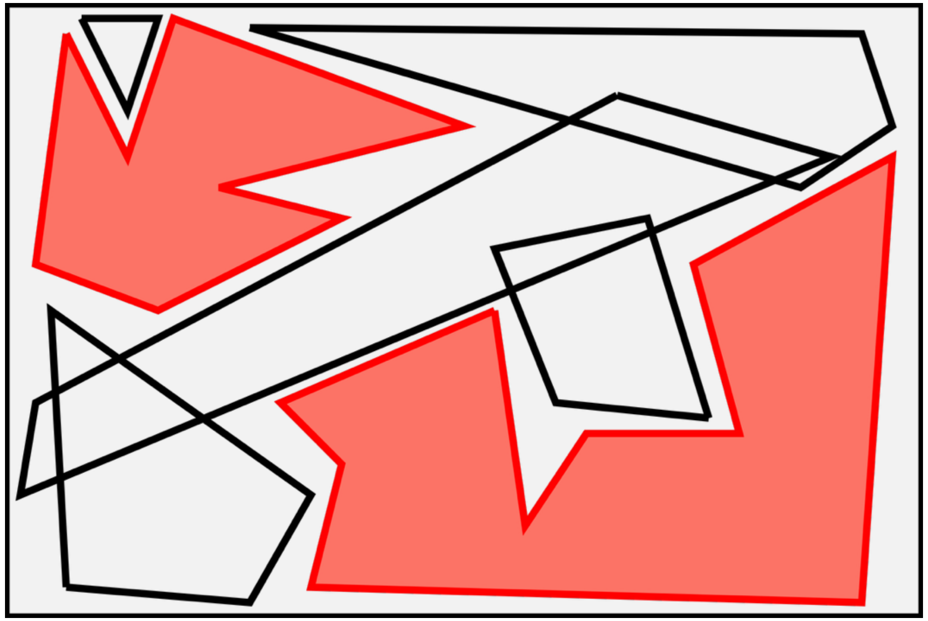


- How can we describe C-free?

1. **Sampling-Based Methods**
    - Probabilistic guarantees of non-collision
    - Probabilistic guarantees of completeness
    - Not inherently optimal

2. **Convex Decomposition**
    - Certificates of non-collision
    - <u>Optimal and complete within the convex decomposition</u>

Collision-Free Configuration Space (C-free)



2015 *IRIS*: Unrealistic requirements on obstacle convexity [1]

2023 *C-IRIS*: Provides guarantees of non-collision [2]

2023 *IRIS-NP*: Faster and larger regions than C-IRIS, but probabilistic guarantees of non-collision [3]

2024 *IRIS-NP2* and *IRIS-ZO*: Fastest variants and larger polytopes, use sampling to accelerate [4]

2025 *EI-ZO:* Regions fast enough for online motion planning [5]

These methods have allowed for the first time to formulate global motion planning as an optimisation problem in relatively high-dimensions, with strong guarantees of non-collision

From Motion to Manipulation Planning

Motion Planning

- Recent advancements allow for optimisation-based, collision-free motion planning within the convex decompositions of C-free
- Has not been tested on manipulation scenarios yet

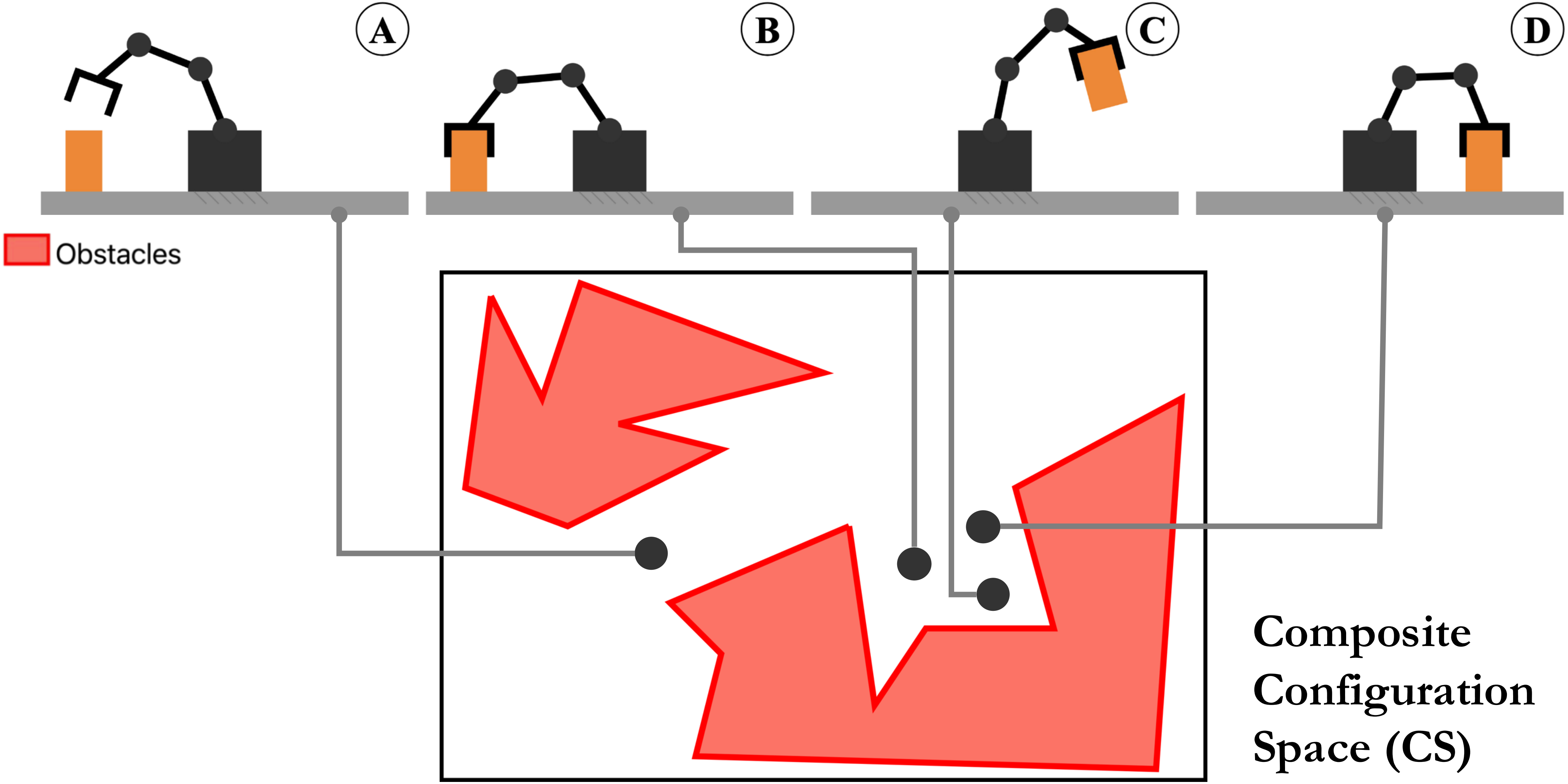
Manipulation Planning

- Usually Sampling-Based approaches
- Cannot guarantee completeness, optimality, and non-collision.
- Needs to strategise about multiple grasping actions

Our Contribution

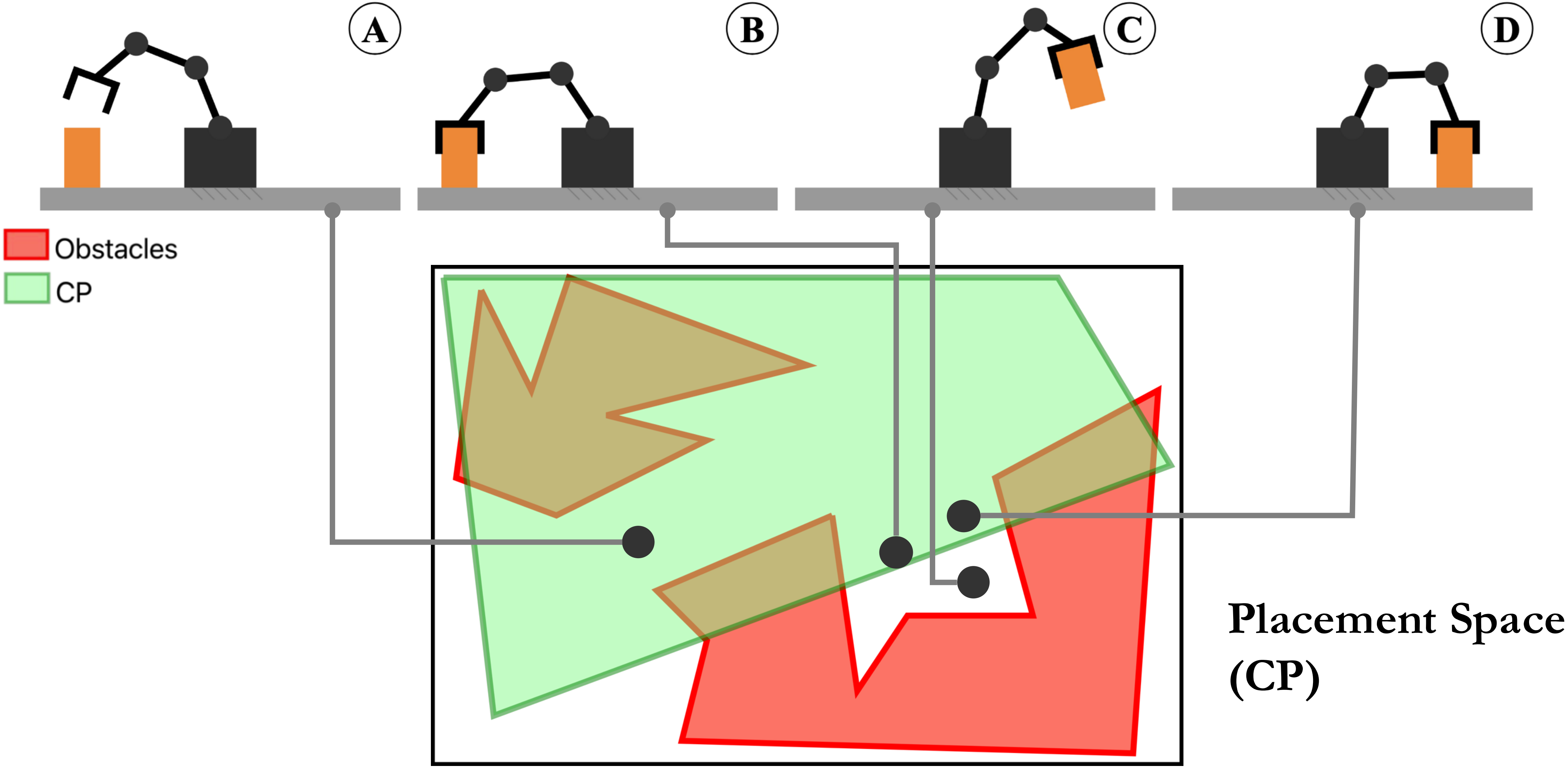
- Formulate a Manipulation Planner that introduces the recent advancements in Convex Decomposition to allow the search of optimal global trajectories

Manipulation Planning as Constrained Motion Planning



Composite Configuration Space (CS)

Manipulation Planning as Constrained Motion Planning



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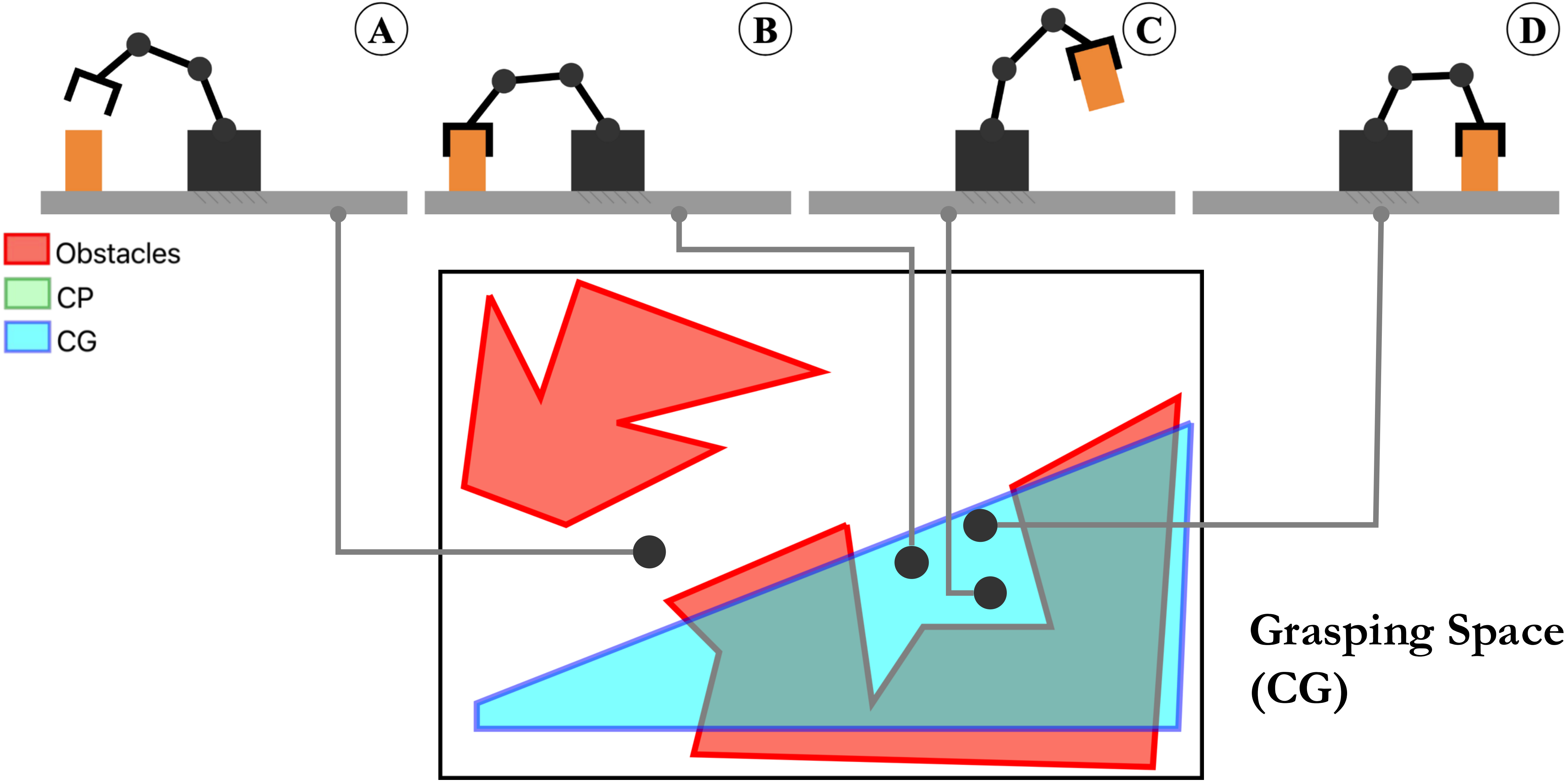

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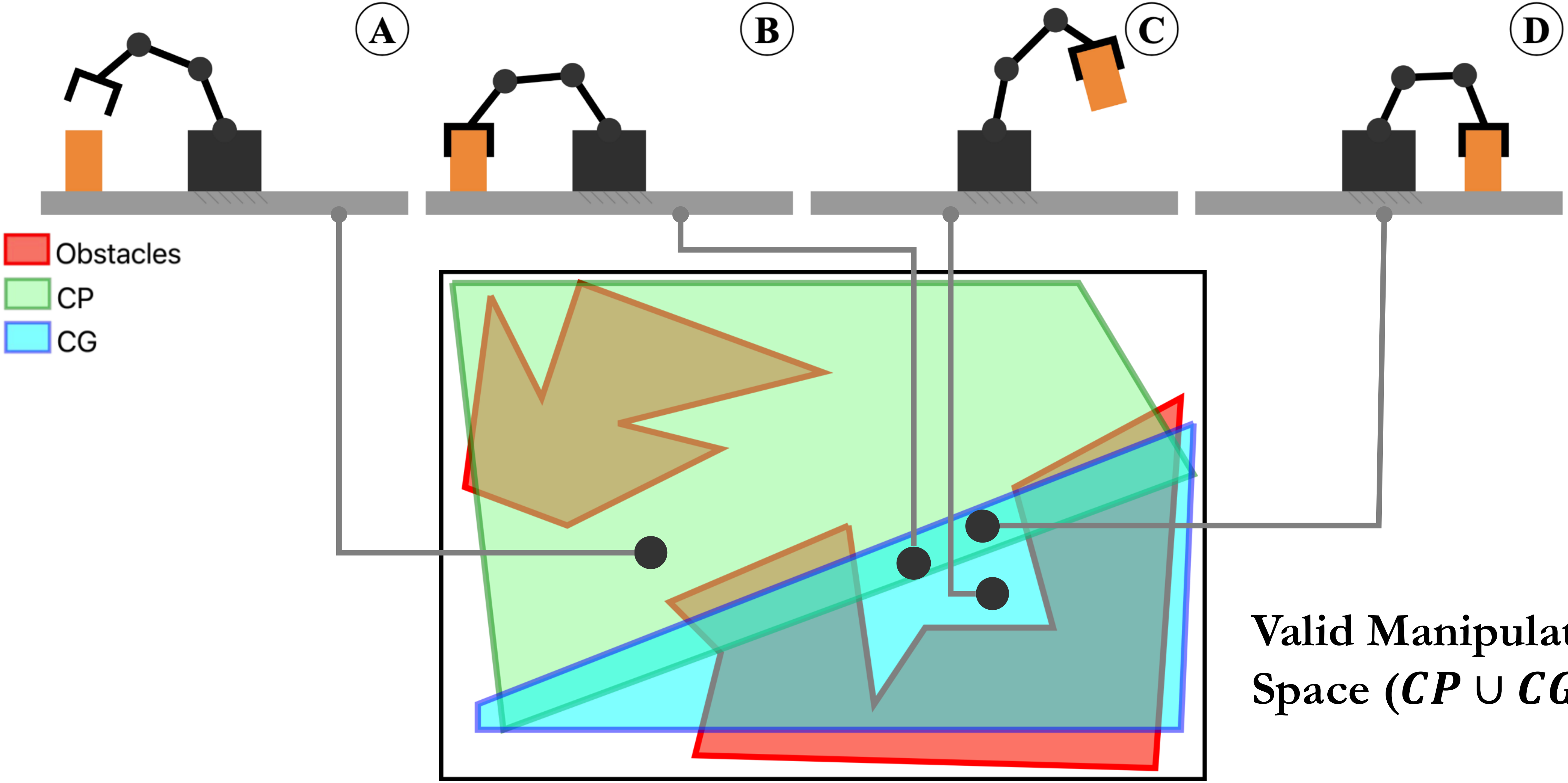
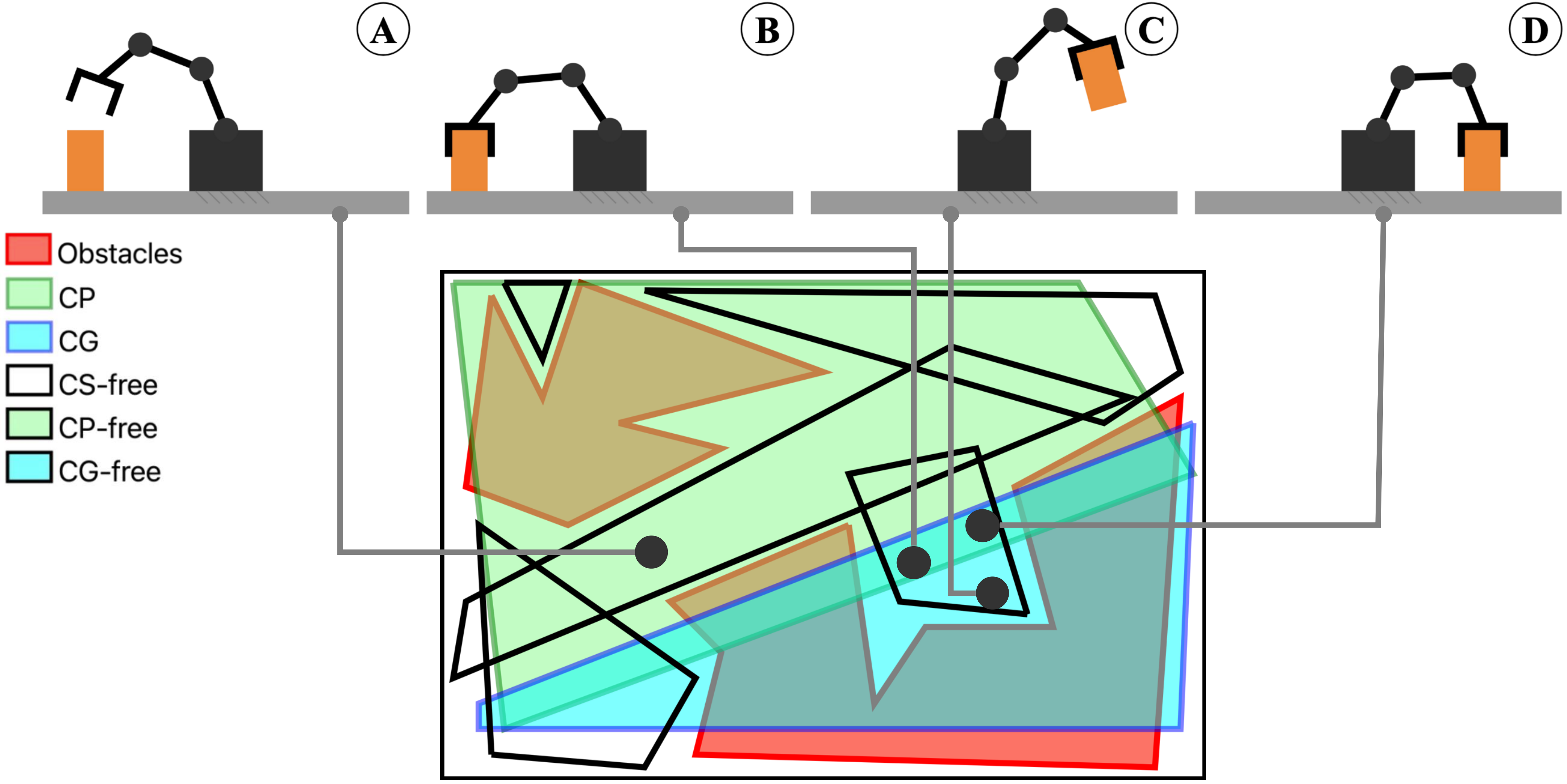


Valid Manipulation Space ($CP \cup CG$)

Manipulation Planning as Constrained Motion Planning



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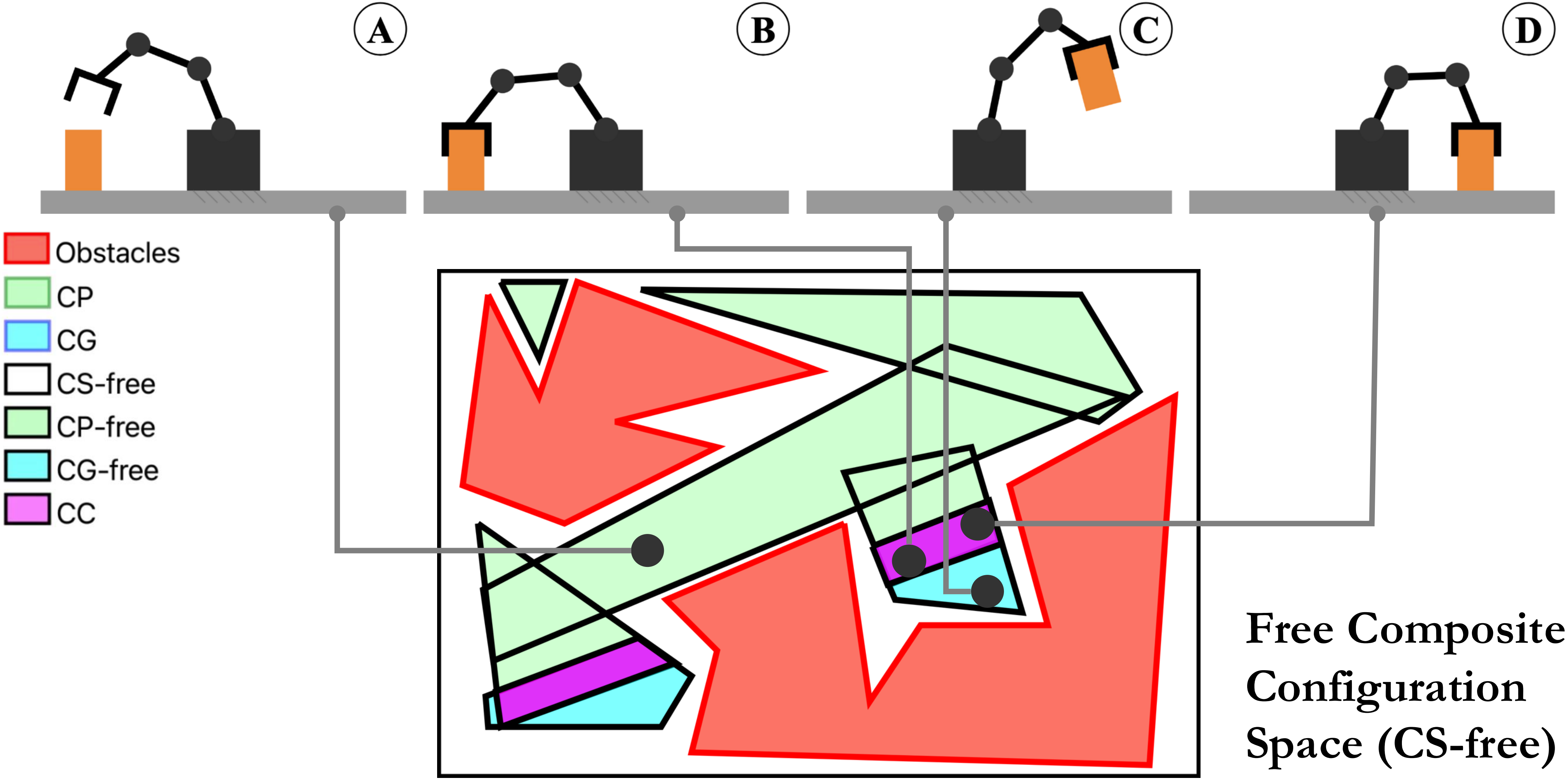

Free Composite Configuration Space (CS-free)

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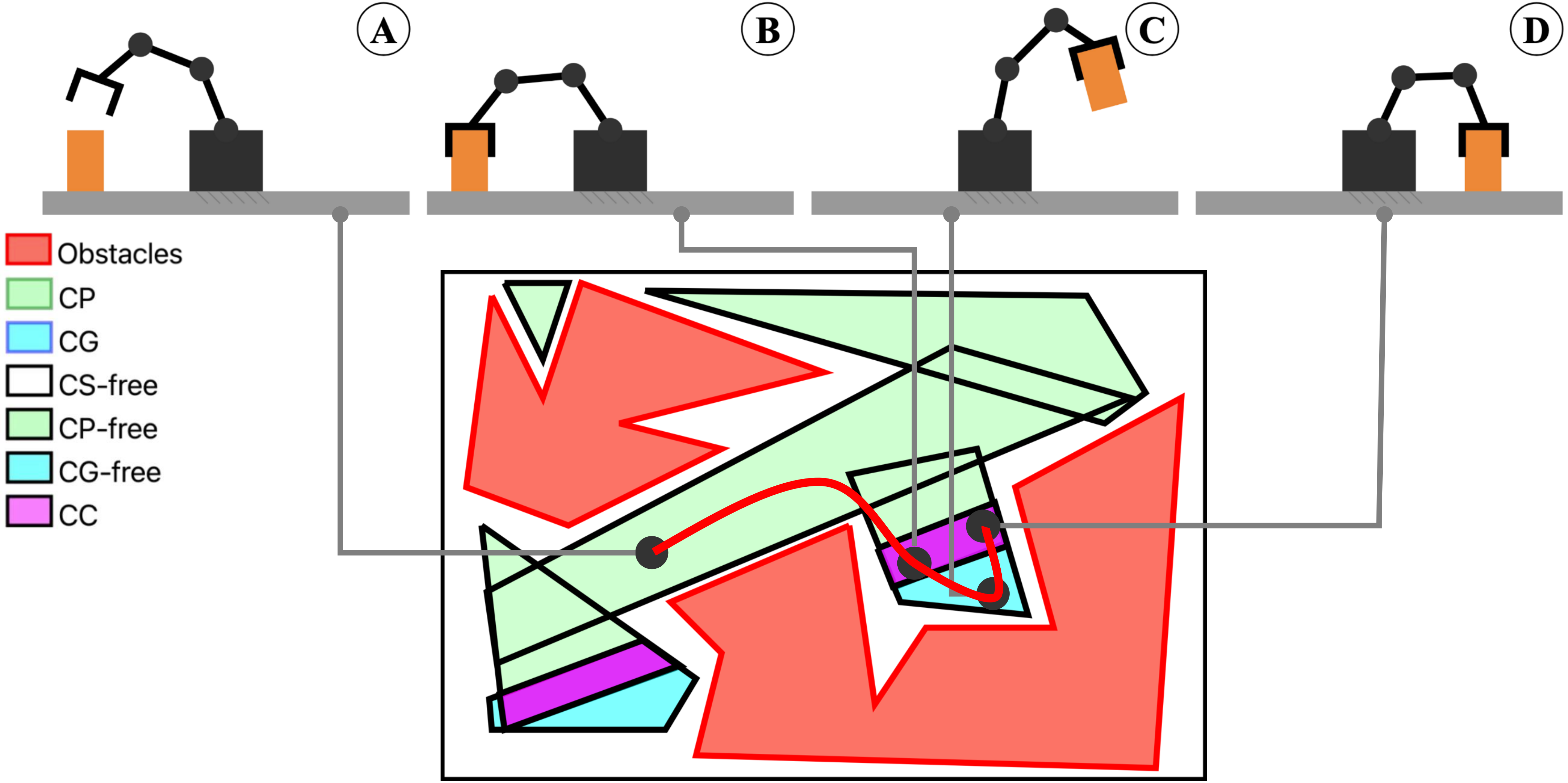

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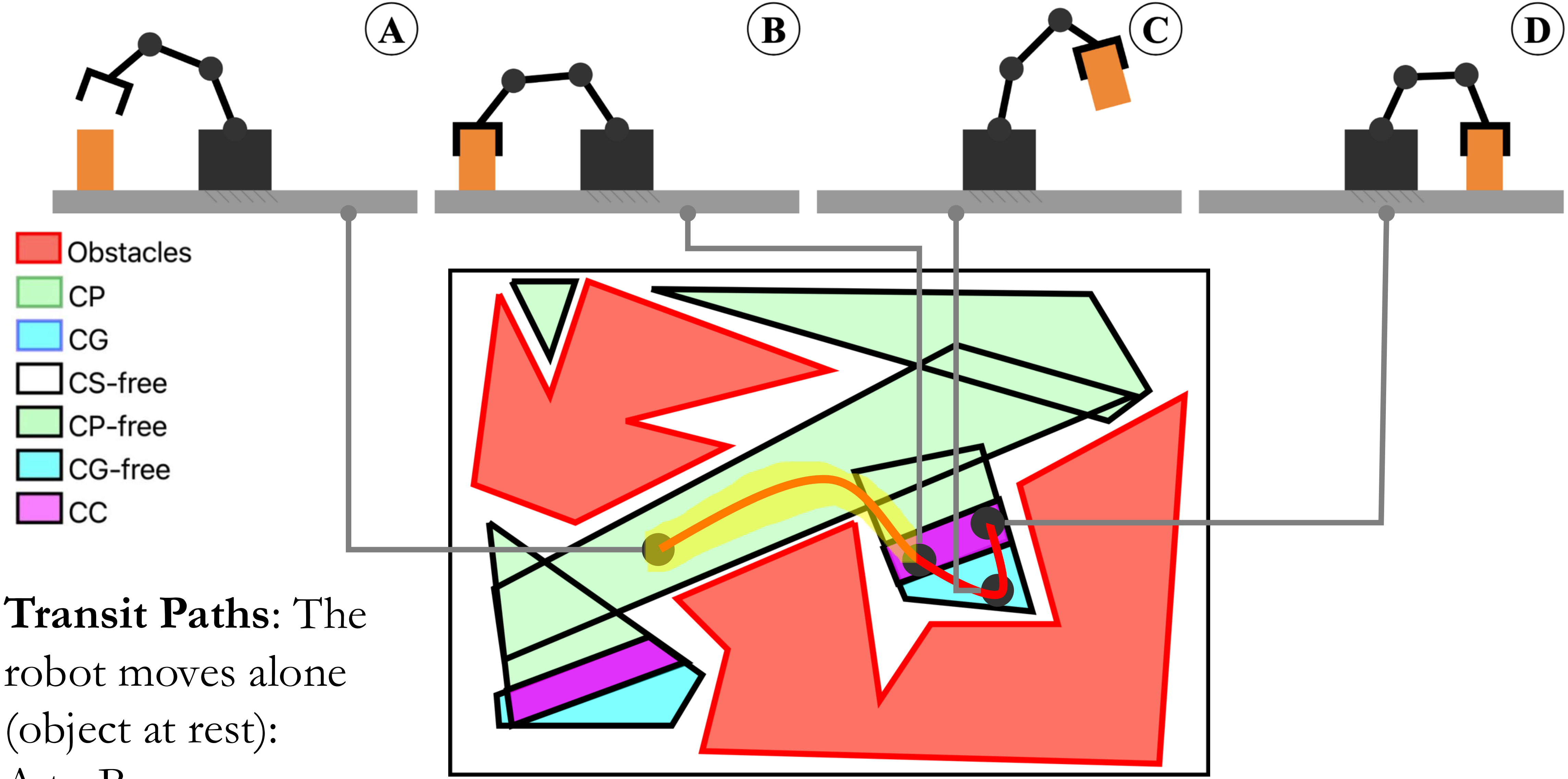
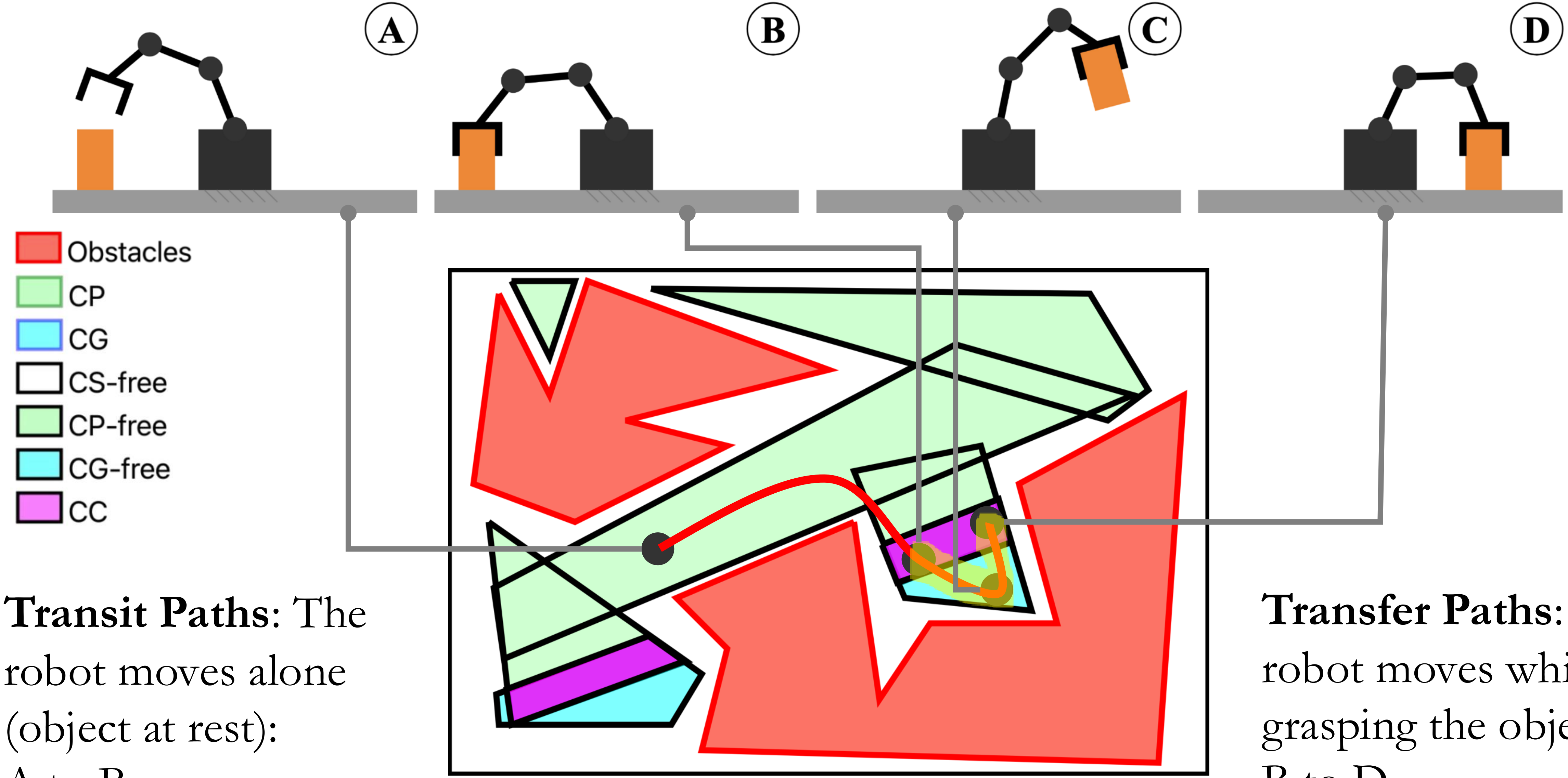


Transit Paths: The robot moves alone (object at rest): A to B

Manipulation Planning as Constrained Motion Planning



Transit Paths: The robot moves alone (object at rest): A to B

Transfer Paths: The robot moves while grasping the object: B to D

Our Optimal Global Manipulation Planner

1. Find Grasp-Release actions that solve the Manipulation Task

(actions that are both in placement and grasping space at the same time)

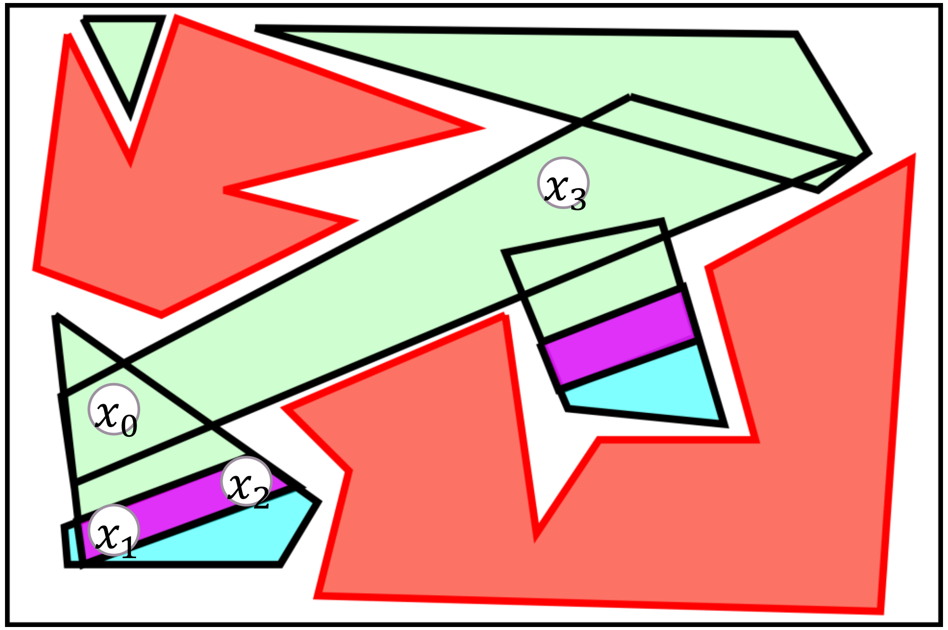
$$\min_{\mathbf{x}_0,\ldots,\mathbf{x}_{2\ell+1}} \sum_{i=0}^{2\ell} \|\mathbf{x}_{i+1} - \mathbf{x}_i\|_2^2$$

$$\text{s.t.} \quad \mathbf{q}_{\min} \leq \mathbf{x}_i \leq \mathbf{q}_{\max}$$

Transfer Constraints

Transit Constraints

Connectivity Constraints



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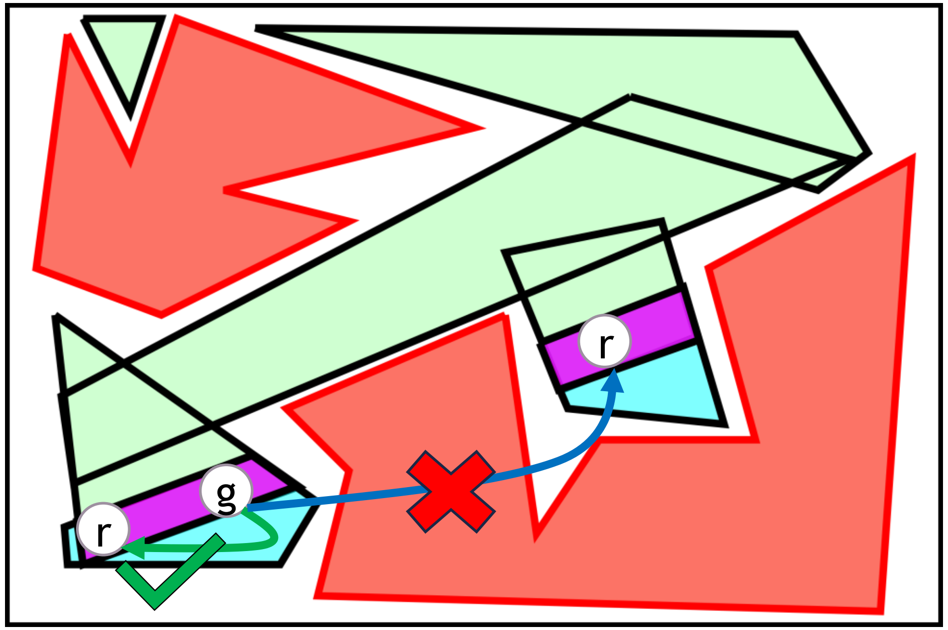
Reachability → Connectivity Constraints



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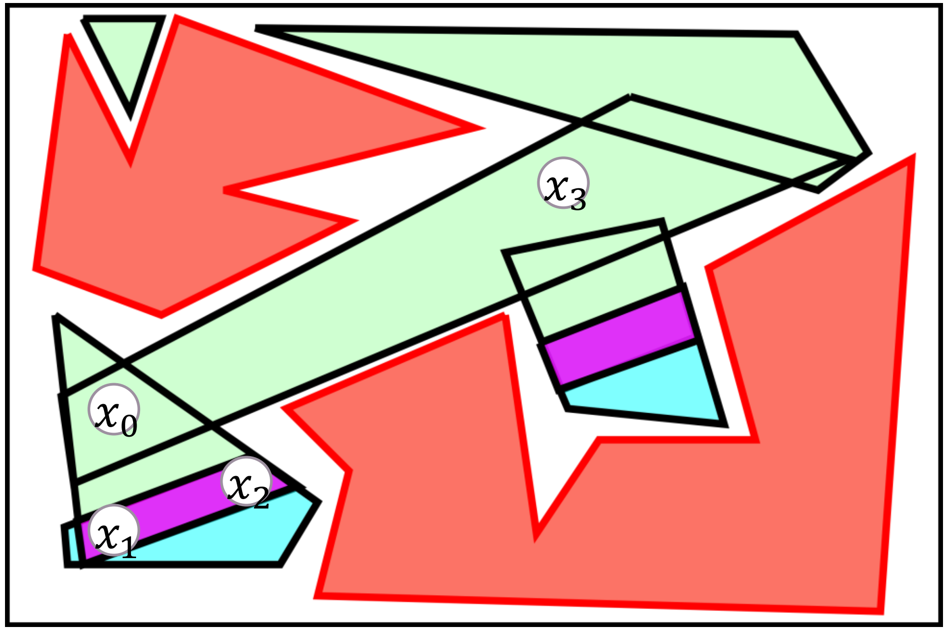
Feasibility



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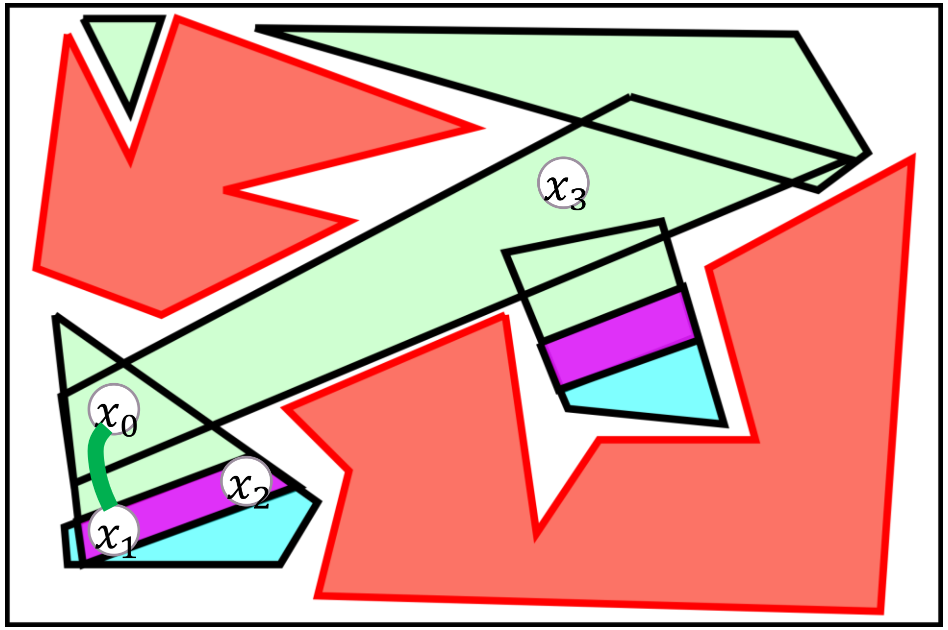

2. **Find the transit and transfer trajectories that connect the grasp-release states**

$$\tau = \bigcup_{i=0}^{2\ell} \tau_i \quad \text{where} \quad \tau_i = \mathrm{GCS}(\mathbf{x}_i, \mathbf{x}_{i+1}),$$

[6]

$$\tau_i = \begin{cases} \text{Transit Path} & \text{if } i \text{ is even} \\ \text{Transfer Path} & \text{if } i \text{ is odd} \end{cases}$$

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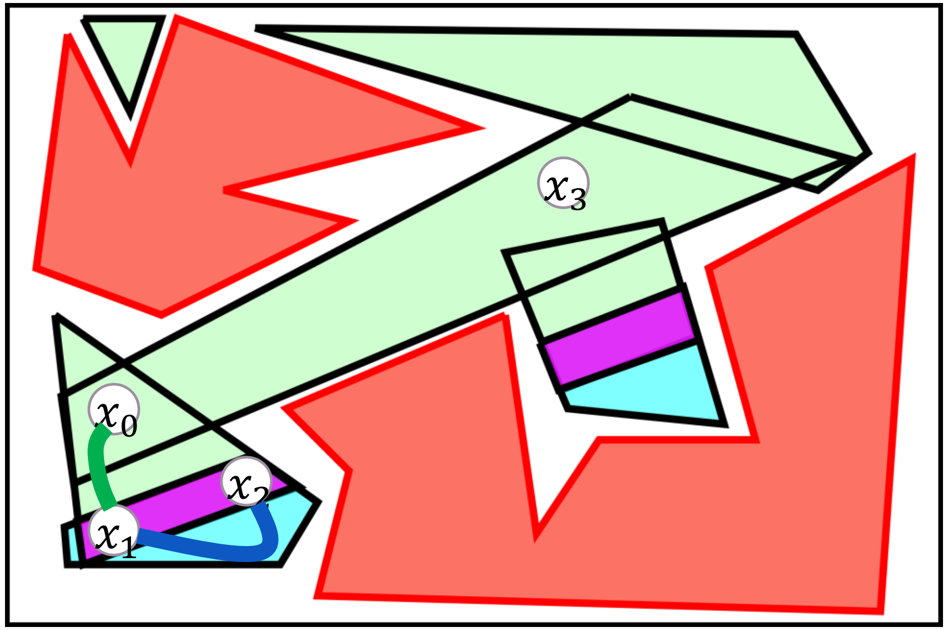

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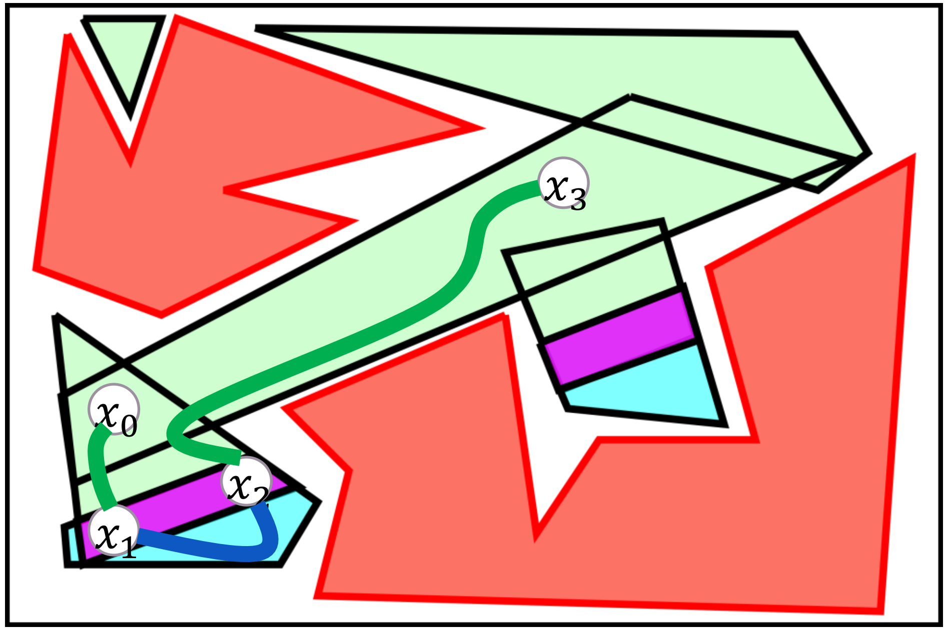

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Our Manipulation Planner Algorithm

Algorithm 1 MANIPULATIONTRAJECTORYPLANNER

Input: Obstacles $\mathcal{O}$, Object $\mathcal{M}$, Robot $\mathcal{R}$, Constraints $\mathcal{C}$, Grasp space CG, Placement space CP, Initial and goal configs $\mathbf{q}_{\text{init}}$ and $\mathbf{q}_{\text{goal}}$, Max grasps K, Certification flag `certify`

Output: Trajectory τ

$CS_{\text{free}} \leftarrow$ DECOMPOSECSPACEWITHCLIQUECOVER$(\mathcal{O}, \mathcal{M}, \mathcal{R})$
if `certify` **then**
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$CG_{\text{free}}, CP_{\text{free}} \leftarrow$ COMPUTEGRASPPLACEMENTFREEREGIONS$(CG, CP, CS_{\text{free}})$
for $\ell = 1$ **to** K **do**
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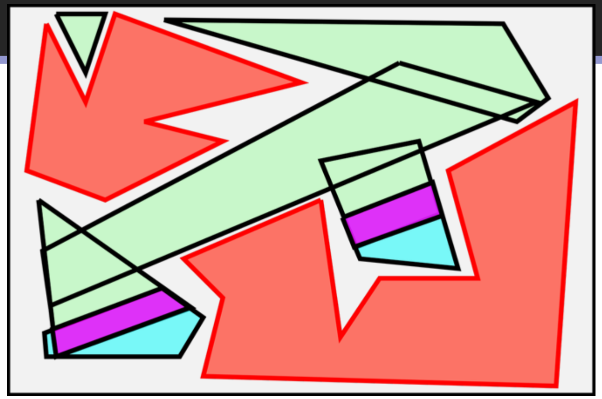
    if $\pi \neq \emptyset$ **then**
        break
    end if
end for
if $\pi = \emptyset$ **then**
    return ”No valid manipulation path found for less than K grasps”
end if
$\tau \leftarrow$ GENERATETRANSFERTRANSITPATHSWITHGCS$(\pi, CS_{\text{free}}, \mathcal{C})$
return τ

Our Manipulation Planner Algorithm



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CP and CG definition and Integration with C-free

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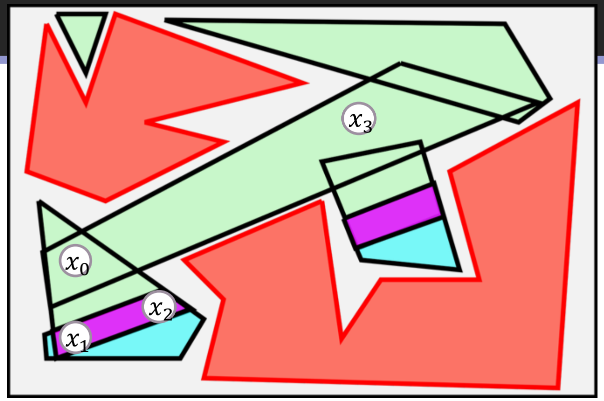

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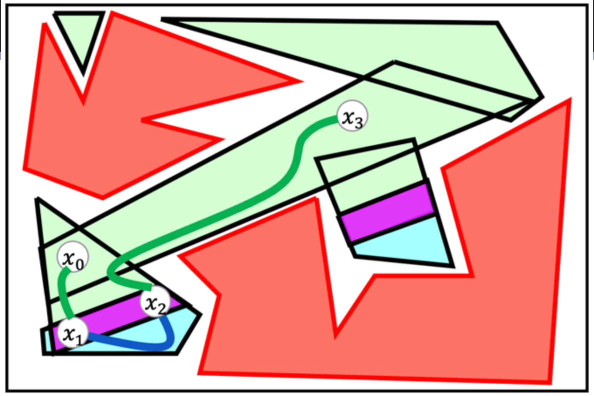
CP and CG definition and Integration with C-free

Finding Grasp and Release actions that solve the task using Mixed-Integer Quadratic Programming

Our Manipulation Planner Algorithm



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Finding Grasp and Release actions that solve the task using Mixed-Integer Quadratic Programming

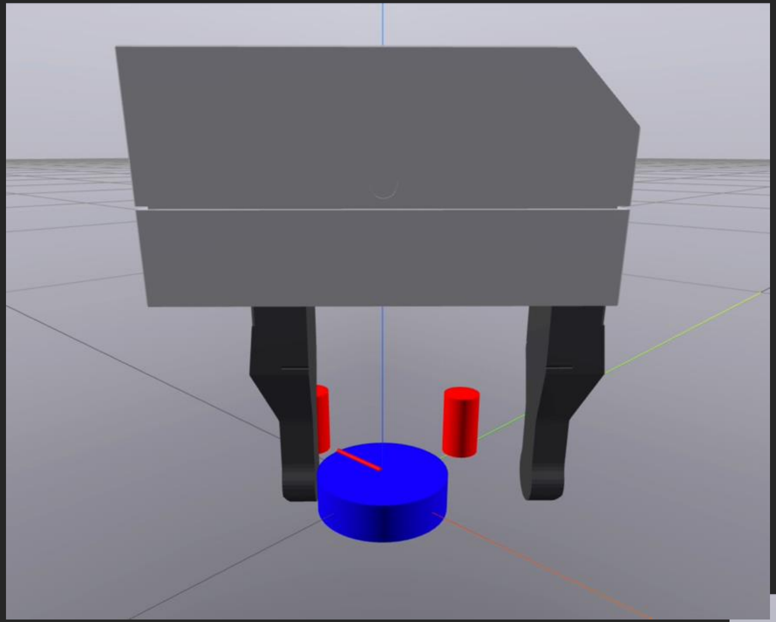
$\tau \leftarrow$ GENERATETRANSFERTRANSITPATHSWITHGCS$(\pi, CS_{\text{free}}, \mathcal{C})$

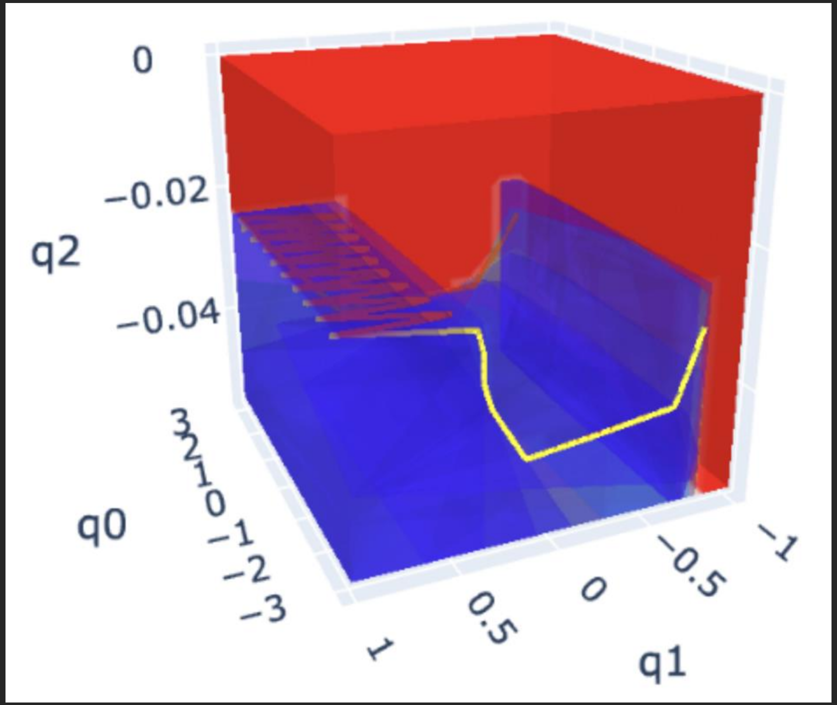
Connect the Grasp and Release actions to form a Trajectory

return τ

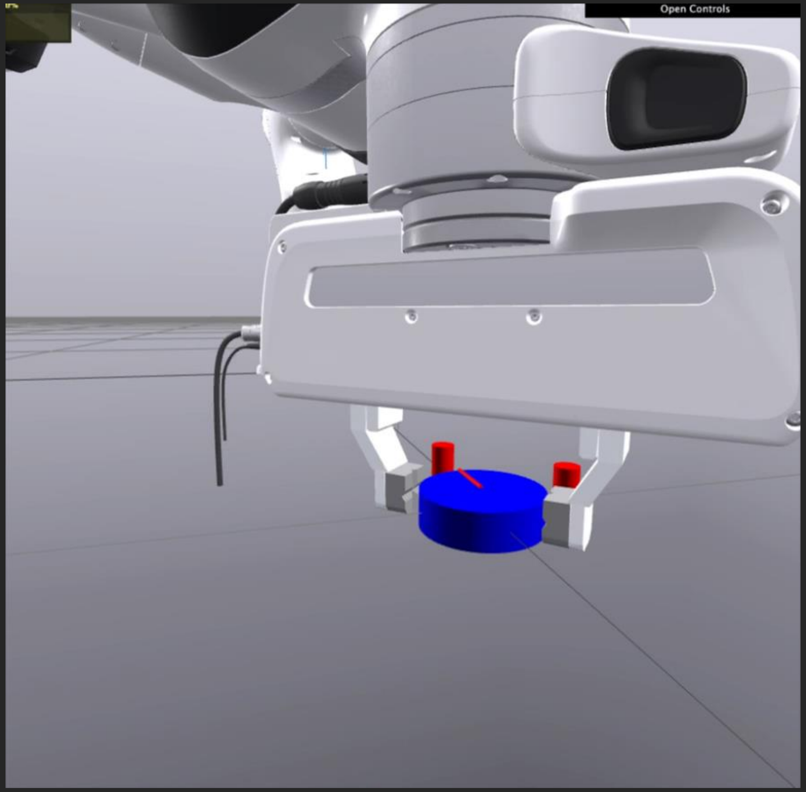
Results

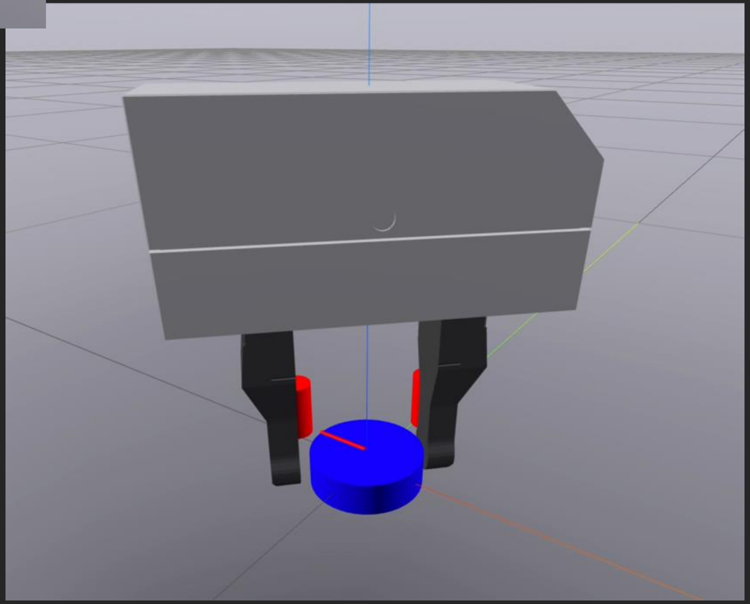
4-DOF





4-DOF in FRANKA





3-DOF

Work in Progress: Higher Dimensional Systems

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So far:

- Convex Decomposition of C-free in higher dimensions
- Manually defined grasp/placement polytopes

Work in Progress: Higher Dimensional Systems

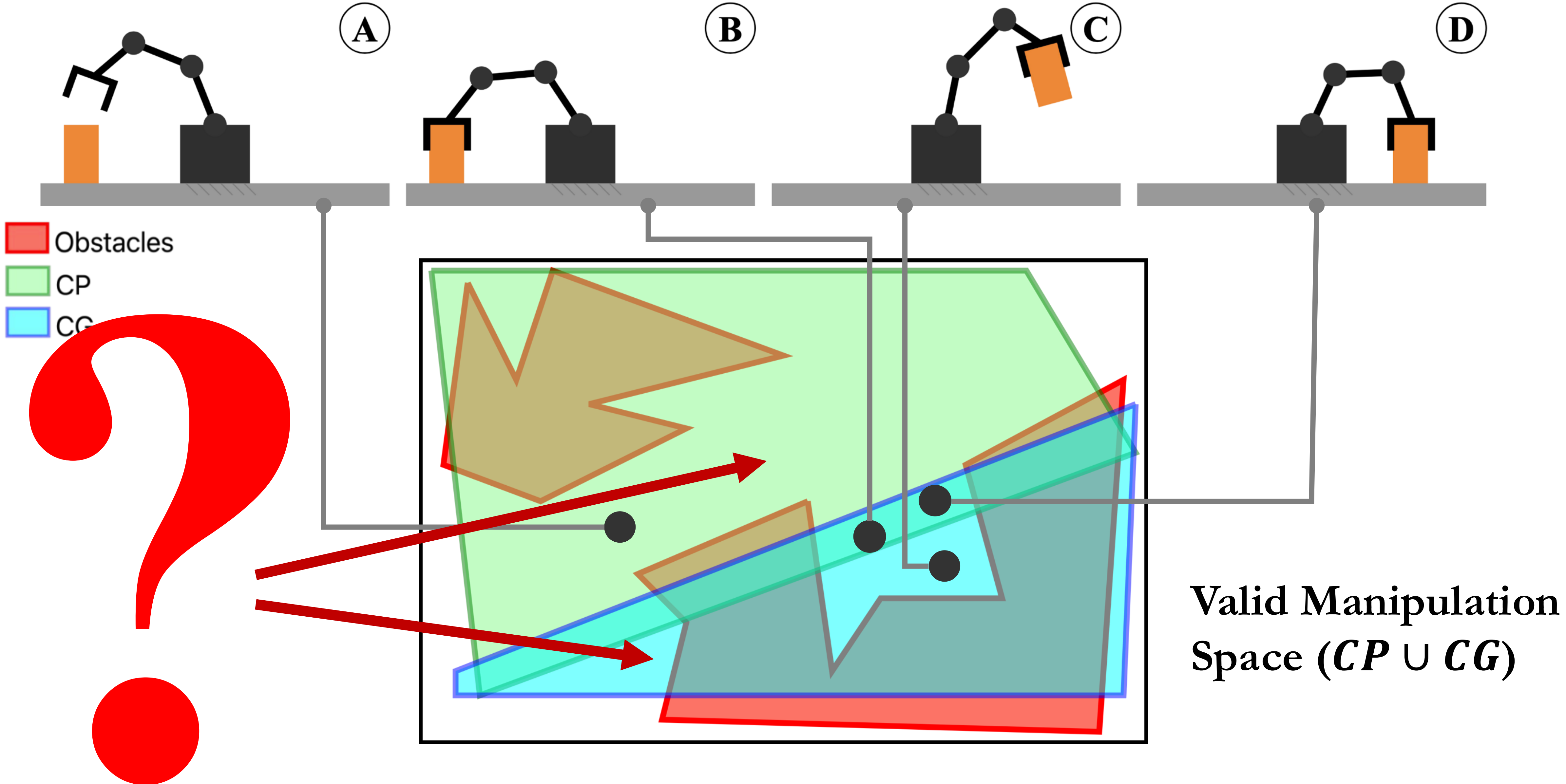
So far:

- Convex Decomposition of C-free in higher dimensions
- Manually defined grasp/placement polytopes

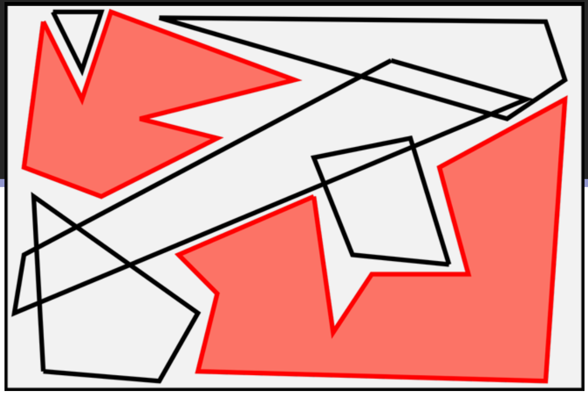
Missing:

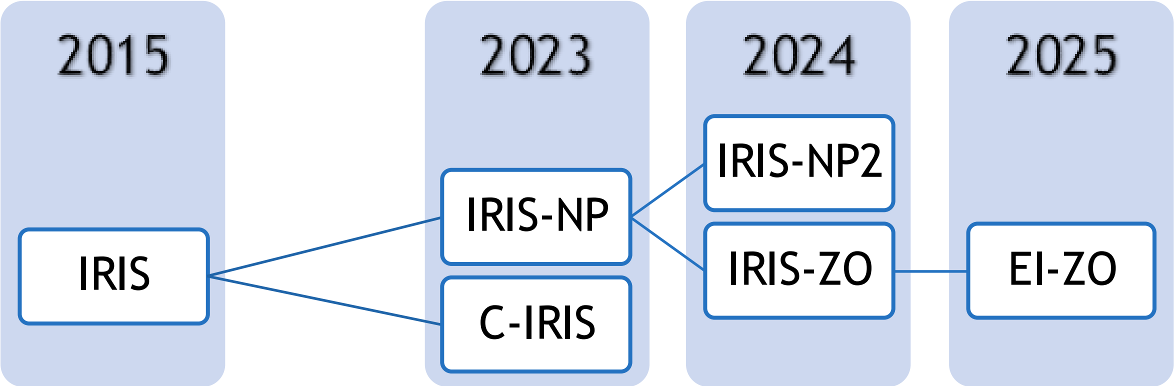
- Convex Decomposition of Grasping and Placement Spaces in higher dimensions

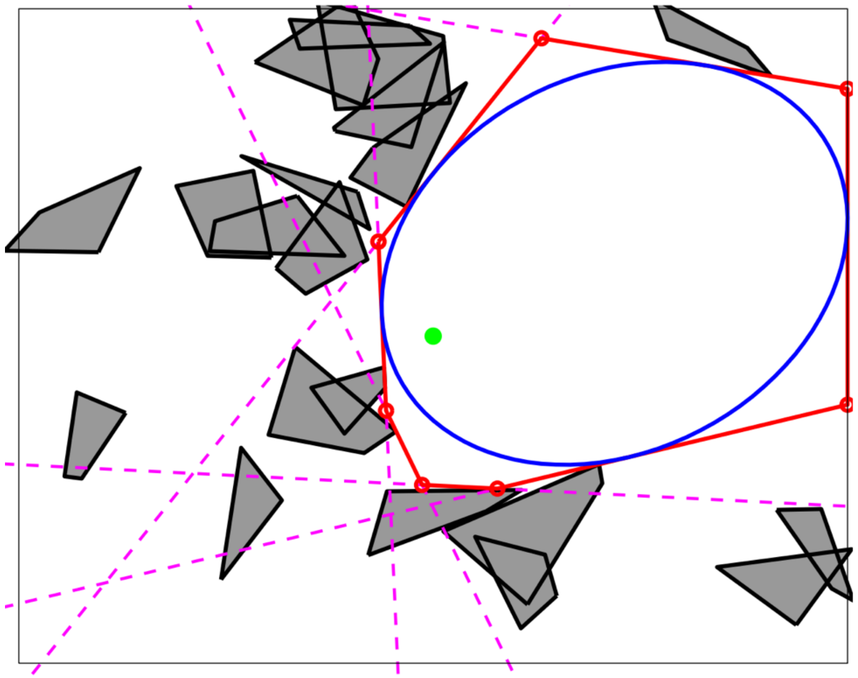
How do we generate the Grasping/Placement polytopes?



C-free Convex Set Generation Methods



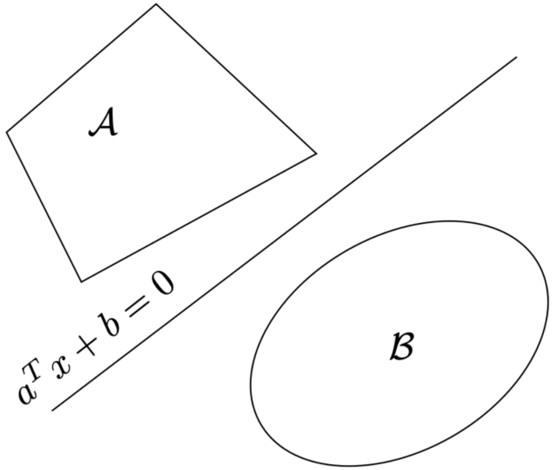




Find $a,\ b$

$$a^T x + b > 0,\ \forall\, x \in \mathcal{A}$$

$$a^T y + b < 0,\ \forall\, y \in \mathcal{B}$$



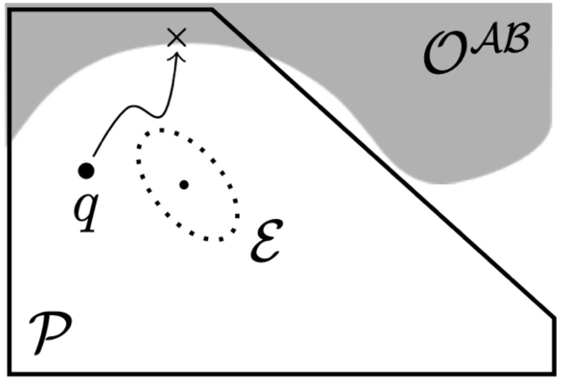
All of these algorithms:

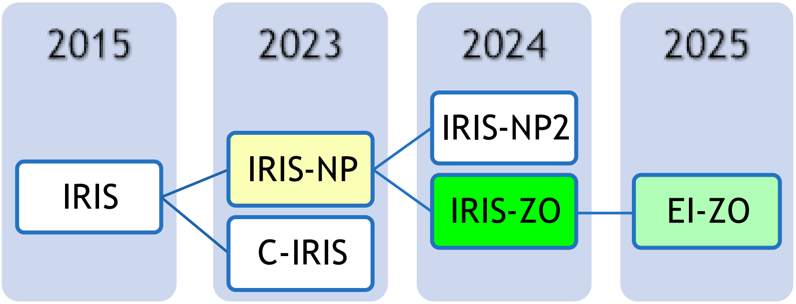
- Find **separating hyperplanes between free region and the collision sets** (obstacles)
- Maximise the volume of the polytope through maximum volume inscribed ellipsoid

C-free Convex Set Generation Methods

IRIS-NP

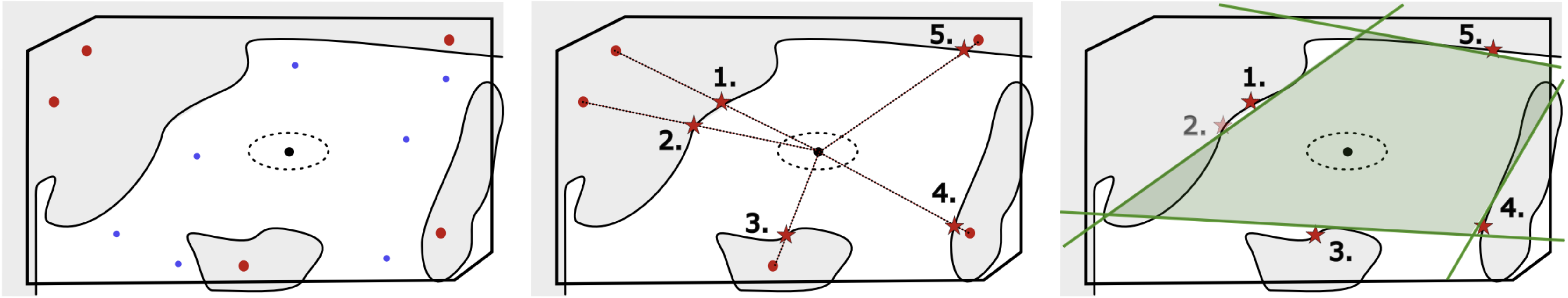
- Using Non-linear Optimisation
- Slow and limited reliability





IRIS-ZO (and EI-ZO)

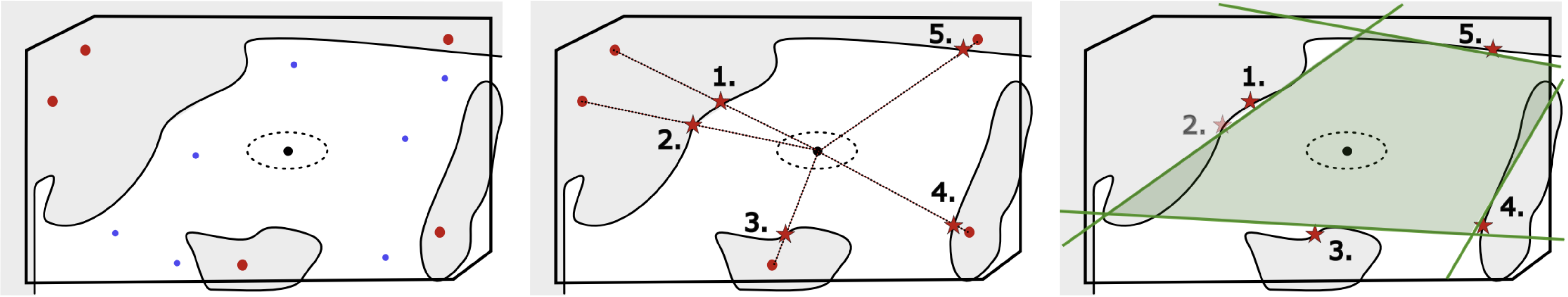
- Zero-Order Optimisation
- Massively Parallelizable with GPU computations



Idea: Modify IRIS-ZO using Valid Grasp Samples

IRIS-ZO

- Zero-Order Optimisation
- Massively Parallelizable with GPU computations

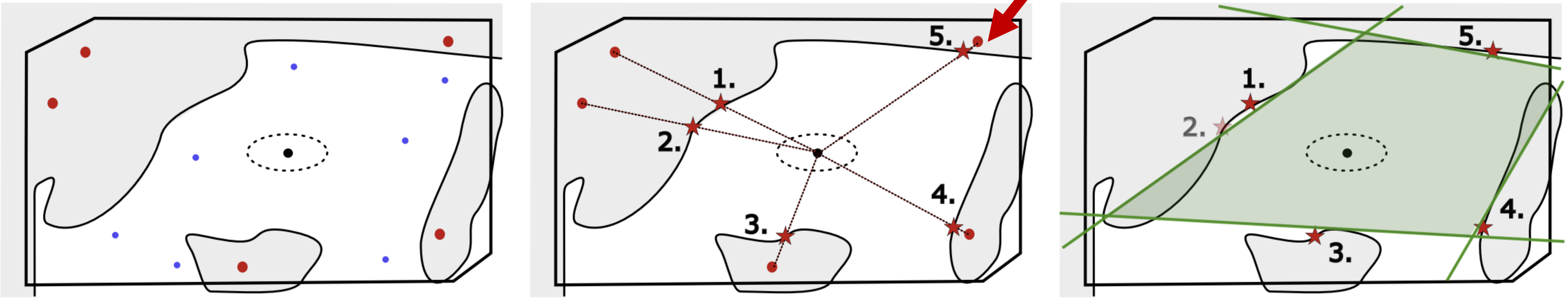


Idea: Modify IRIS-ZO using Valid Grasp Samples

IRIS-ZO

- Zero-Order Optimisation
- Massively Parallelizable with GPU computations

Check for Valid Grasps instead of Collision



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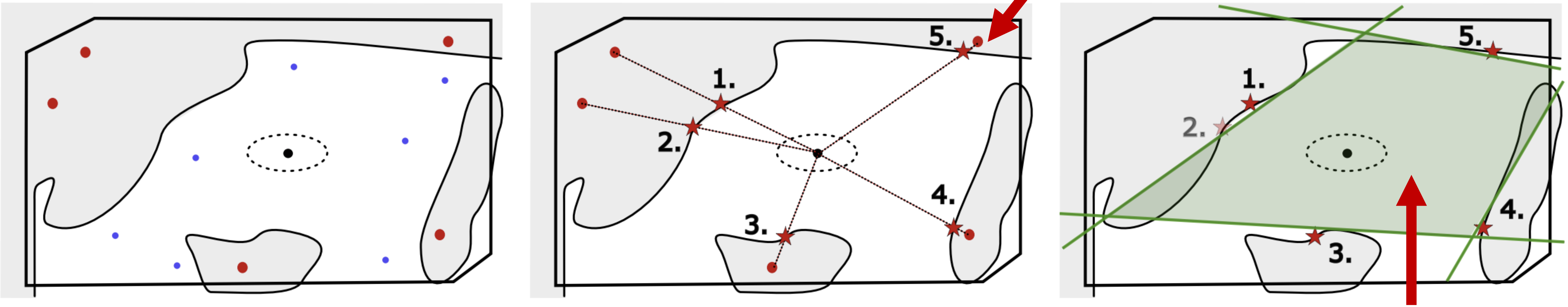

Grasping Polytope

Idea: Modify IRIS-ZO using Valid Grasp Samples

IRIS-ZO

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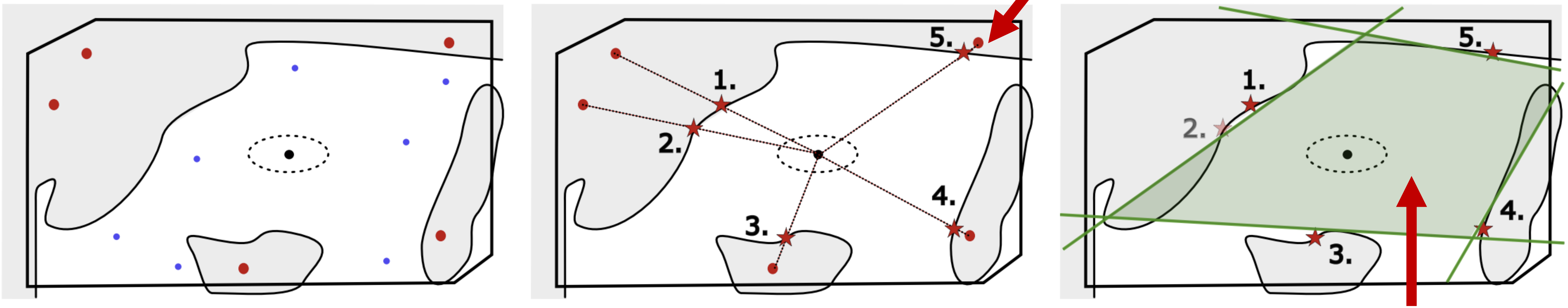
Check for Valid Grasps instead of Collision



Grasping Polytope

Challenge: Grasping space is more difficult to sample *uniformly*

Idea: Modify IRIS-ZO using Valid Grasp Samples

IRIS-ZO

- Zero-Order Optimisation
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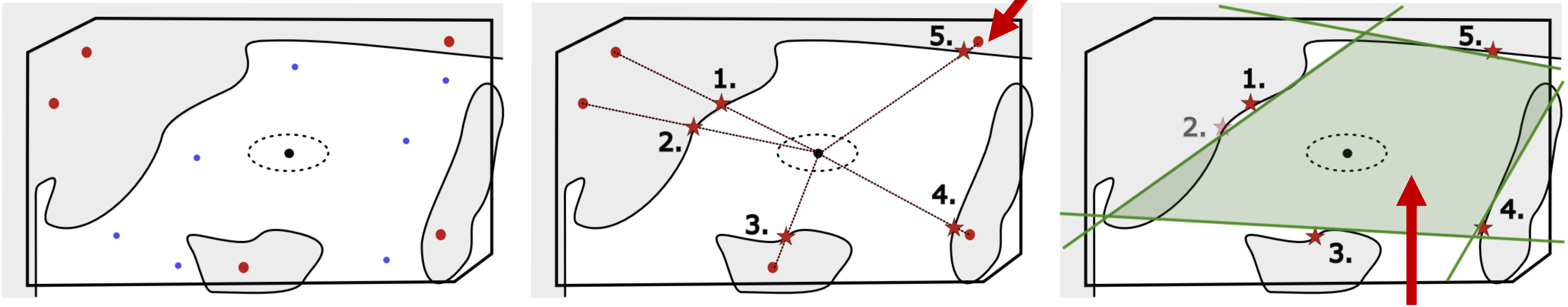


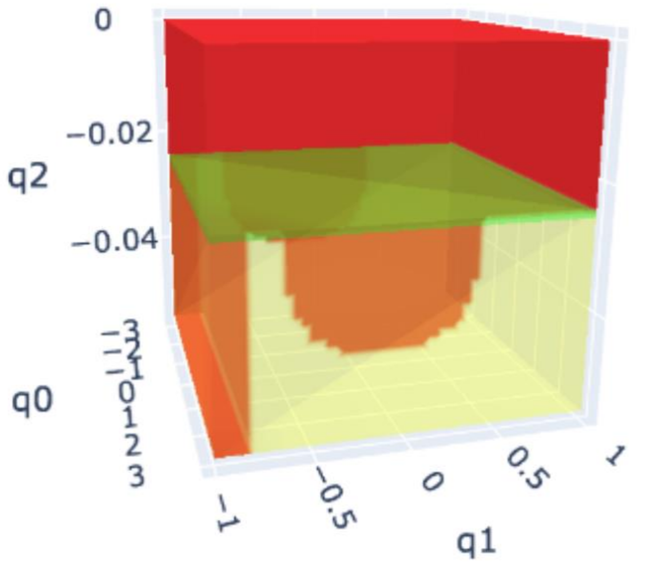
Grasping Polytope

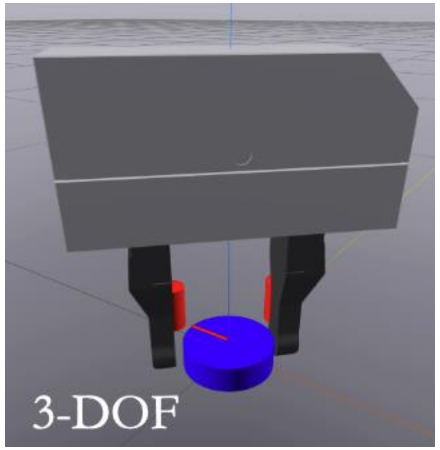
Challenge: Grasping space is more difficult to sample *uniformly*

Our solution: Use advances in NLP Sampling

So far

- Because the gripper was fixed, we could define grasping as linear inequalities on the finger joints (a convex polytope)





- For a full arm, *grasping becomes a nonlinear constraint* on the forward kinematics

So far

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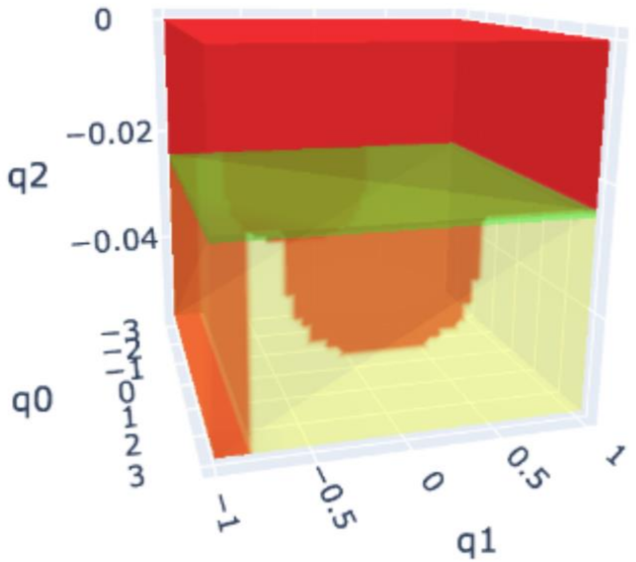
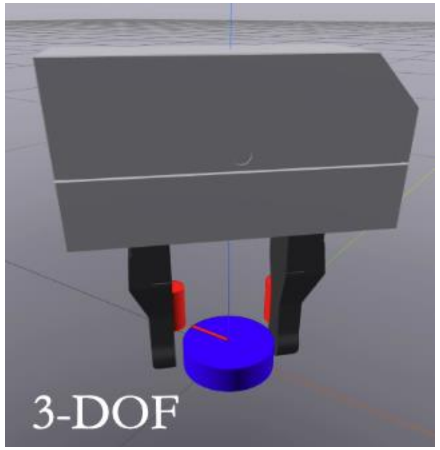
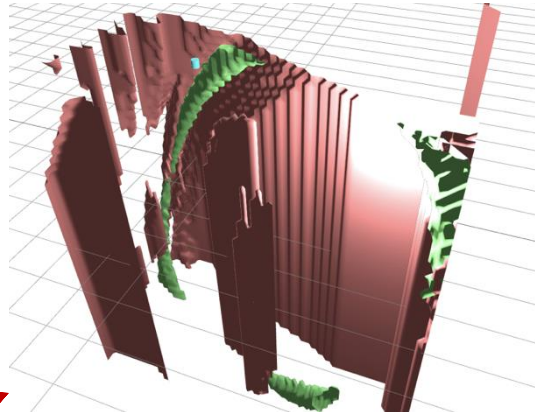




- For a full arm, *grasping becomes a nonlinear constraint* on the forward kinematics

NLP Sampling

- Sampling in constrained spaces is challenging.
- If the constraints are nonlinear, the space is narrow and disconnected: difficult for random exploration



- *Workaround*: sample near the constraint and project to manifold

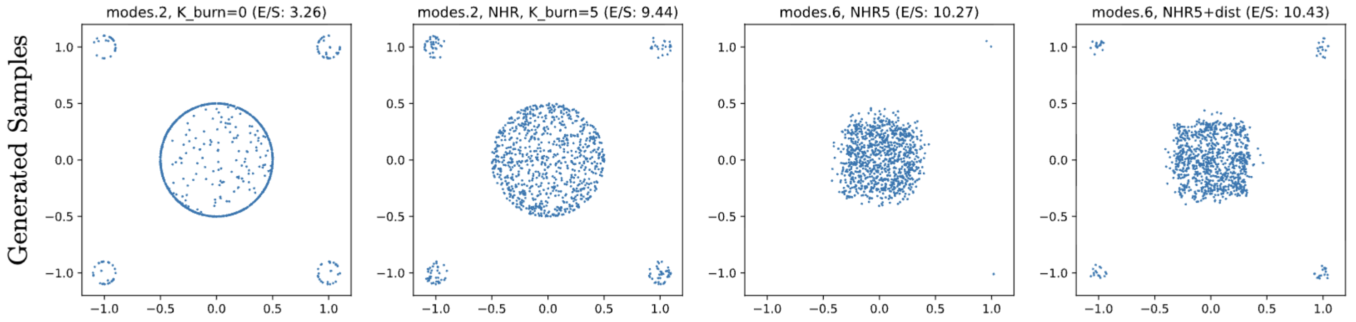
- **Problem:** We need to cover the entire constraint space

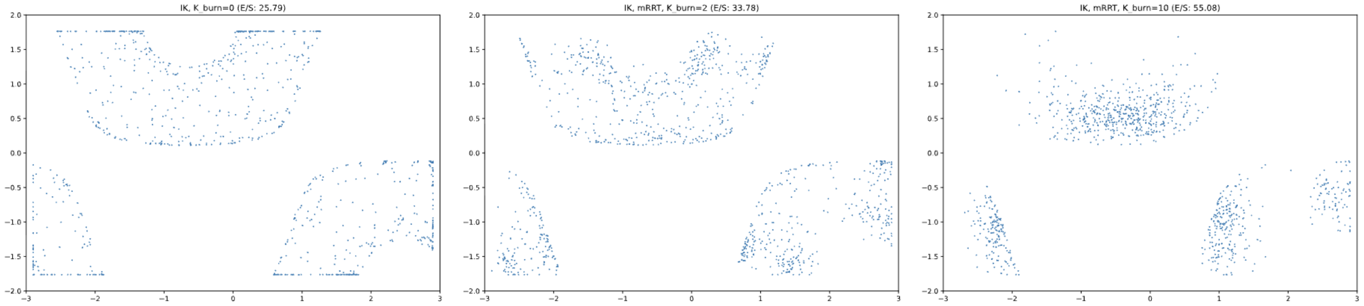
NLP Sampling – Toussaint et al. (2024) [7]

- Toussaint et al. (2024) [7] target ***diverse and dense*** sampling of nonlinear manifolds.
- It handles nonlinear equalities and inequalities.
- This is necessary for probabilistic completeness in manipulation problems.

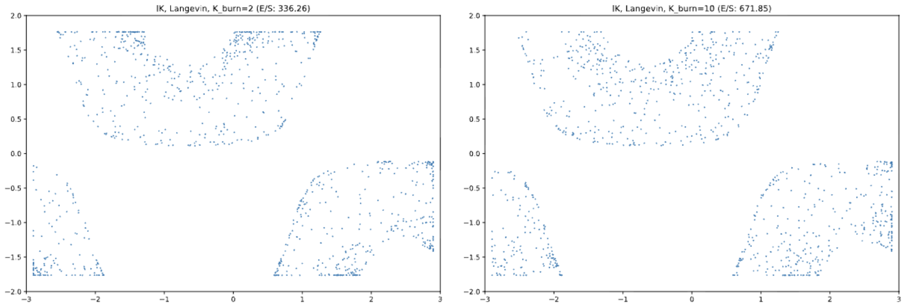
Phase 1: `Find a feasible point`

Phase 2: `Use the found feasible point to seed an interior optimization method.`





(a) Pure GN downhill (b) mRRT with $K_{\text{burn}} = 2$ burn-in steps (c) mRRT with $K_{\text{burn}} = 10$



(d) Langevin with $K_{\text{burn}} = 2$ (e) Langevin with $K_{\text{burn}} = 10$

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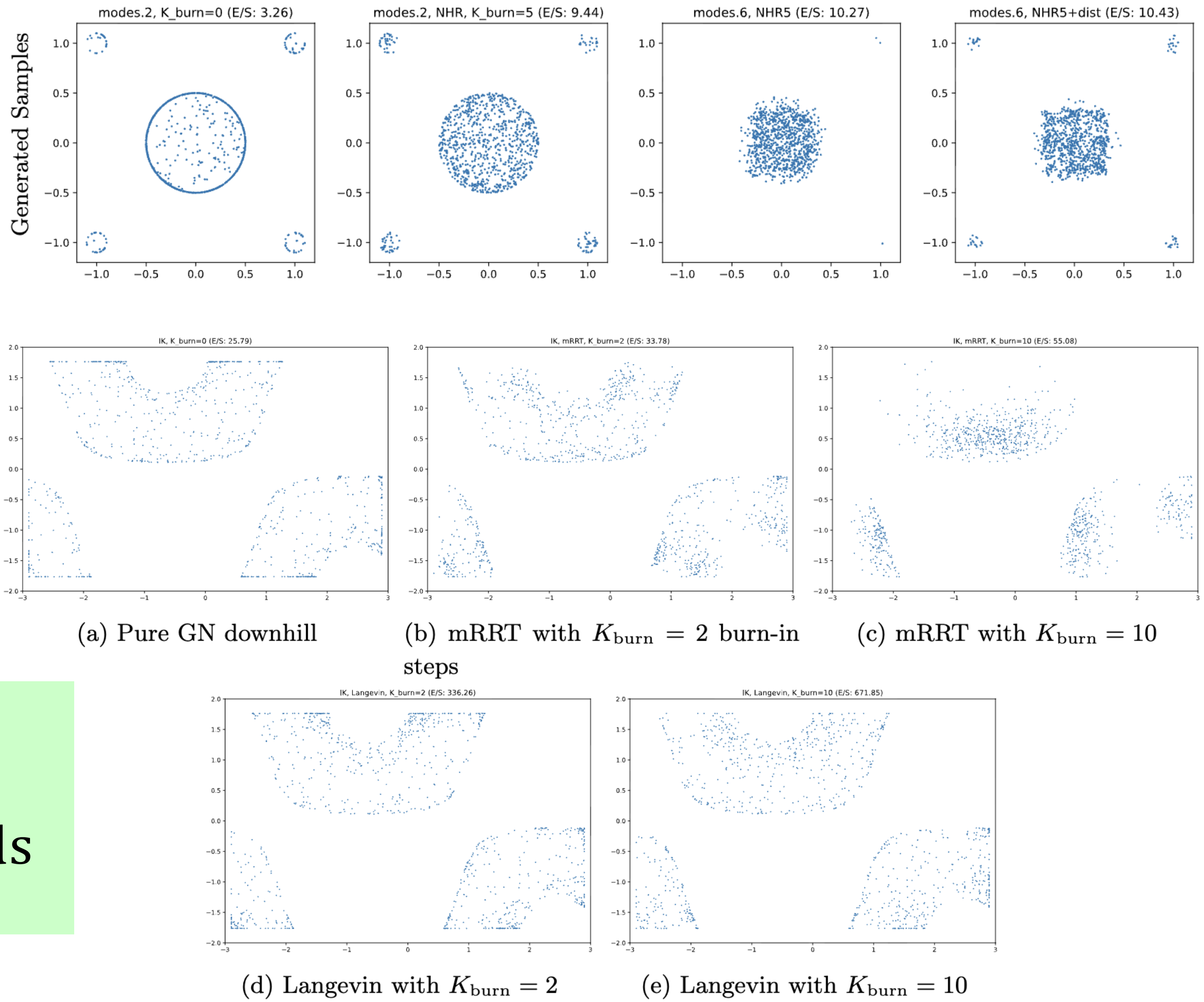

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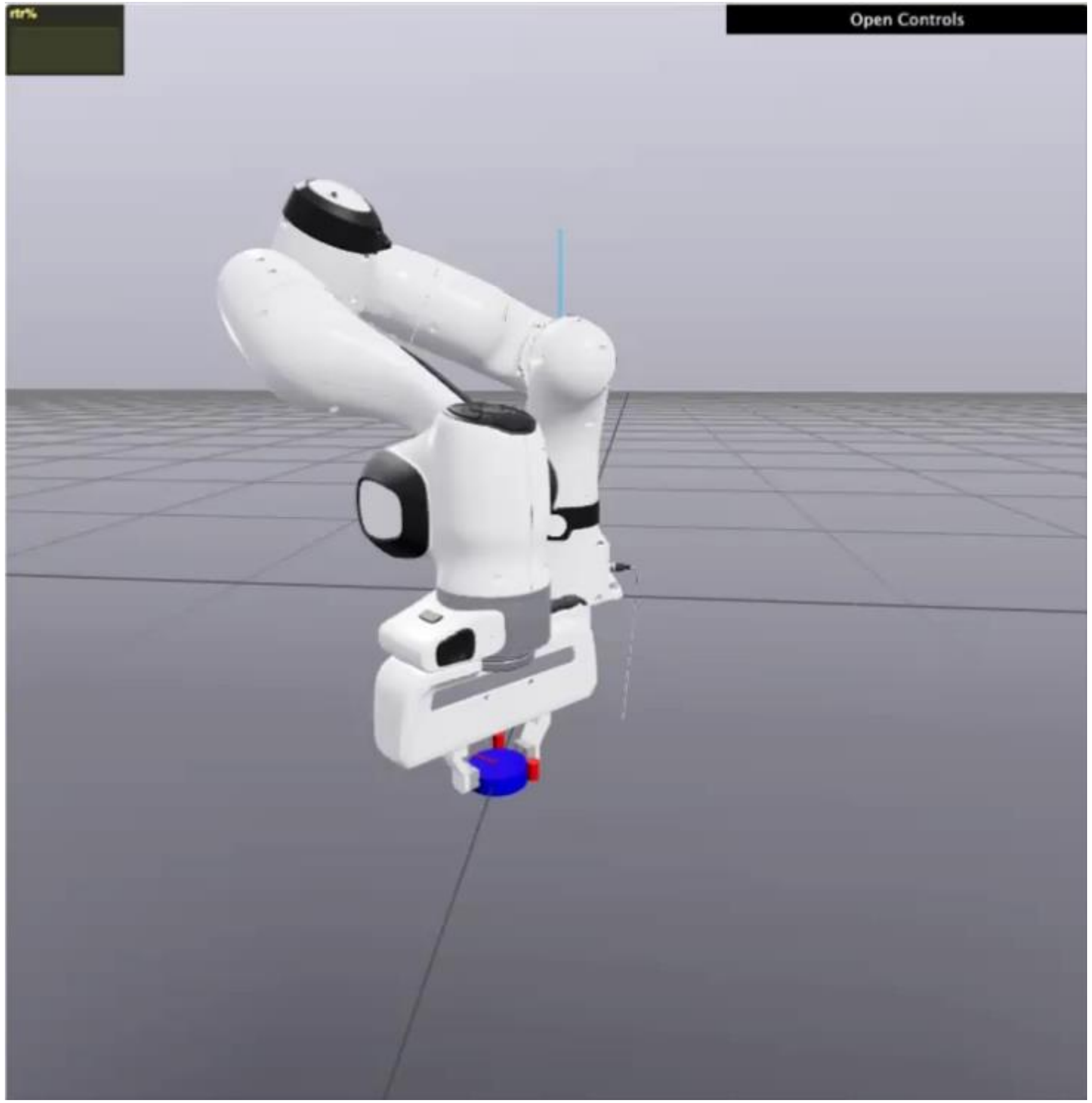
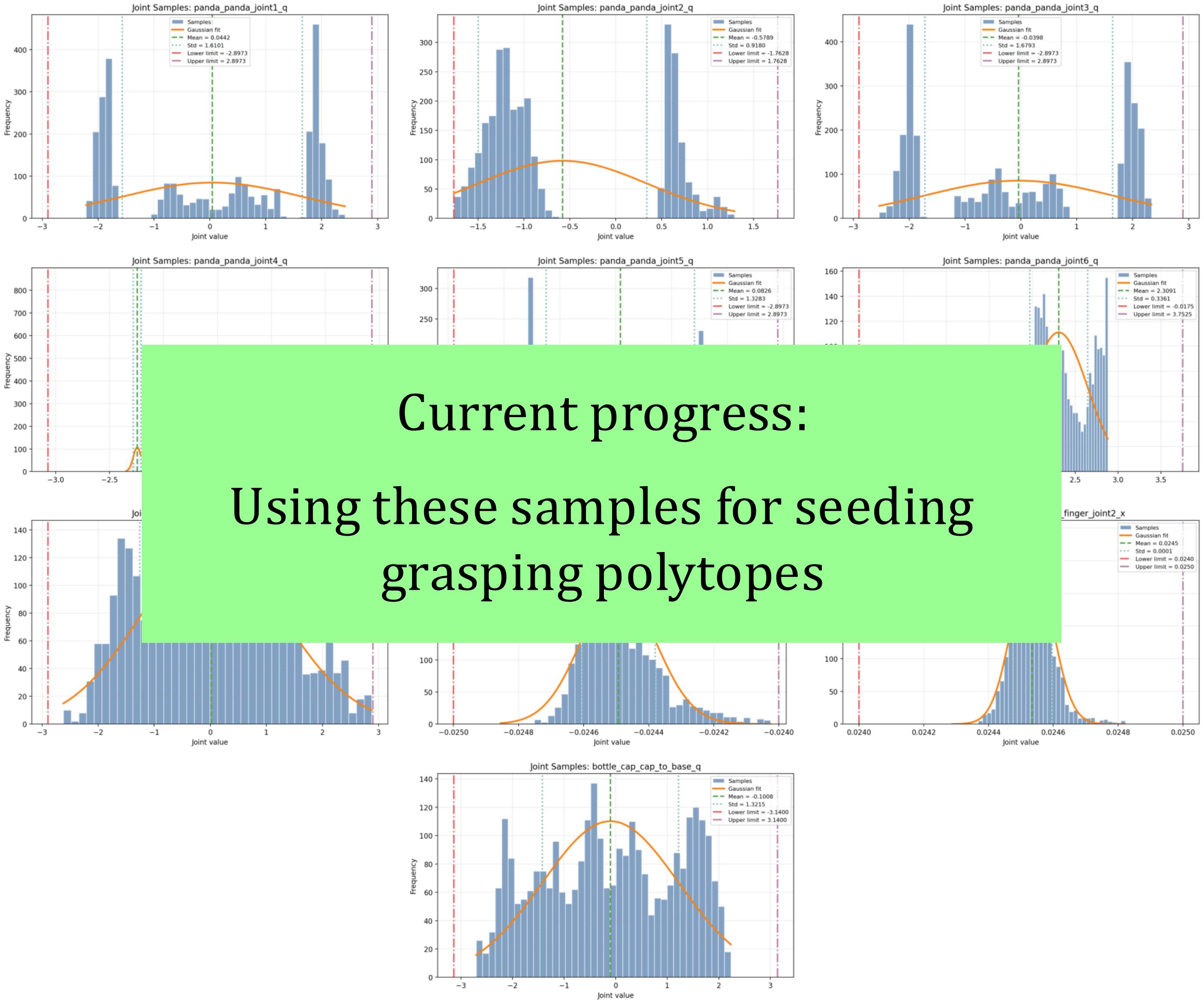
(d) Langevin with $K_{\text{burn}} = 2$ (e) Langevin with $K_{\text{burn}} = 10$

We propose to use these advances for seeding our polytope generation methods

NLP Sampling Results in Grasping Space





Current progress:

Using these samples for seeding grasping polytopes

Our Resulting Manipulation Planner

By constructing the feasible and reachable space of our manipulator, our method offers:

- Completeness and Optimality within the convex representation
- Constraint and Collision-free guarantees
- Global task reasoning with multiple contacts
- Able to quickly generate plans requiring +20 contacts

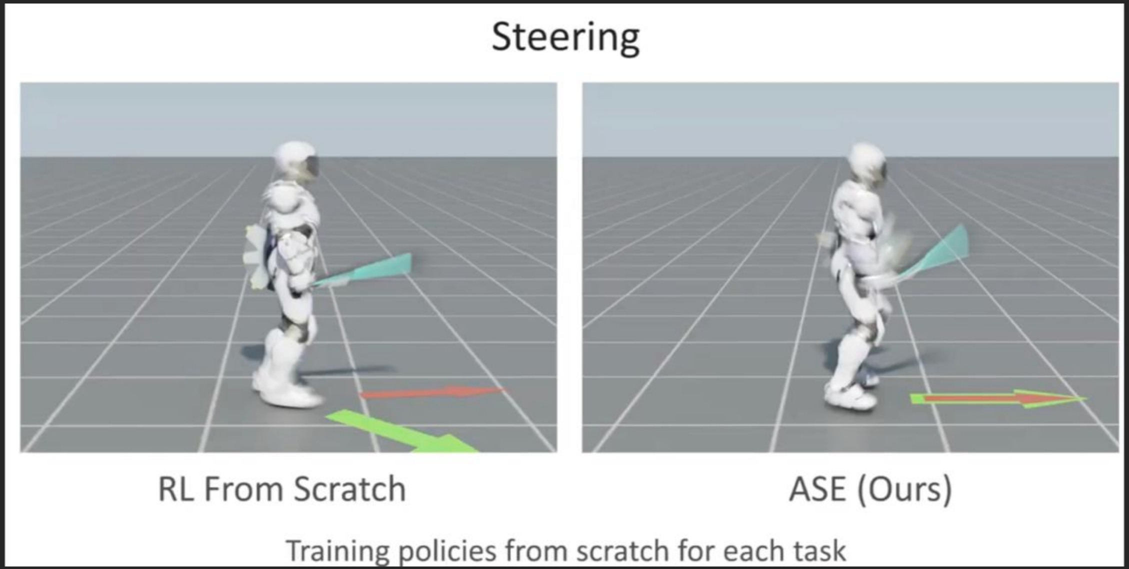
What is missing:

- Dynamics
- Uncertainty
- Partial observability

Next Steps: Where do we go from here?

Planning: Finding a sequence of *actions* that solve a task.

Next step: **Use full motor skills as actions**



[8]

Open question: How do we compose and reuse larger actions to find a global plan?

Questions?

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[1] Deits, R., Tedrake, R. (2015). Computing Large Convex Regions of Obstacle-Free Space Through Semidefinite Programming. In: Akin, H., Amato, N., Isler, V., van der Stappen, A. (eds) Algorithmic Foundations of Robotics XI. Springer Tracts in Advanced Robotics, vol 107. Springer, Cham.

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[3] Petersen, M., & Tedrake, R. (2023). Growing convex collision-free regions in configuration space using nonlinear programming. arXiv preprint arXiv:2303.14737.

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[5] Werner, P., Cheng, R., Stewart, T., Tedrake, R., & Rus, D. (2025). Superfast configuration-space convex set computation on GPUs for online motion planning. arXiv preprint arXiv:2504.10783.

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[7] Toussaint, M., Braun, C. V., & Ortiz-Haro, J. (2024). NLP Sampling: Combining MCMC and NLP Methods for Diverse Constrained Sampling (arXiv:2407.03035).

[8] Peng, X. B., Guo, Y., Halper, L., Levine, S., & Fidler, S. (2022). ASE: Large-Scale Reusable Adversarial Skill Embeddings for Physically Simulated Characters. ACM Transactions on Graphics, 41(4), 1–17.